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Principles of Macroeconomics

Lecture 8: Open-economy macroeconomics

Prof. Dr. Jan-Egbert Sturm

Schedule

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Principles of Macroeconomics

Lecture 7: International trade

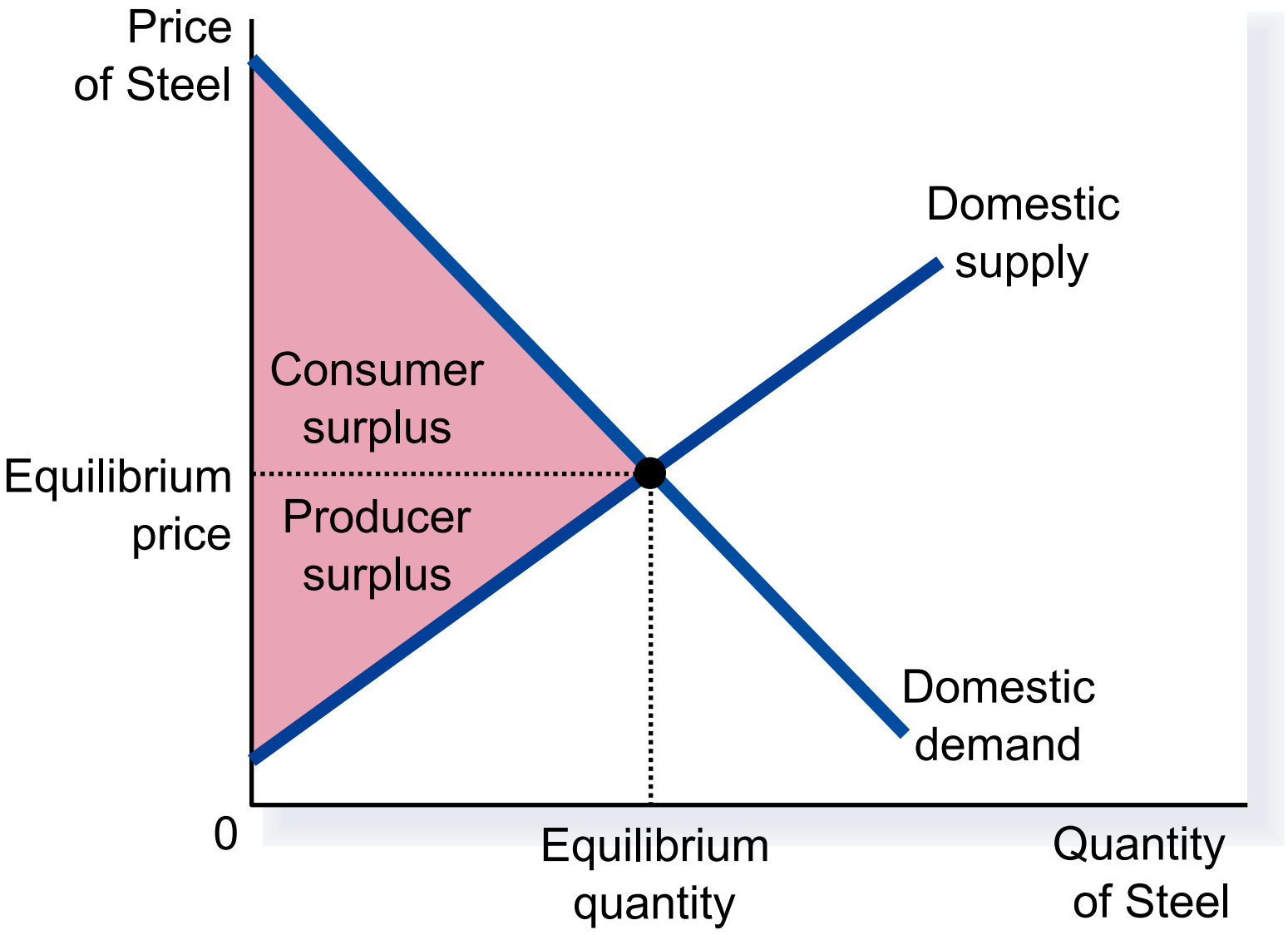
Prof. Dr. Jan-Egbert Sturm

The Determinants of Trade

- Equilibrium Without Trade
  - Assume:
    - A country is isolated from the rest of the world and produces steel
    - The market for steel consists of the buyers and sellers in the country
    - No one in the country is allowed to import or export steel
- Results:
  - Domestic price adjusts to balance demand and supply
  - The sum of consumer and producer surplus measures the total benefits that buyers and sellers receive

- If the country decides to engage in international trade, will it be an importer or exporter of a good?

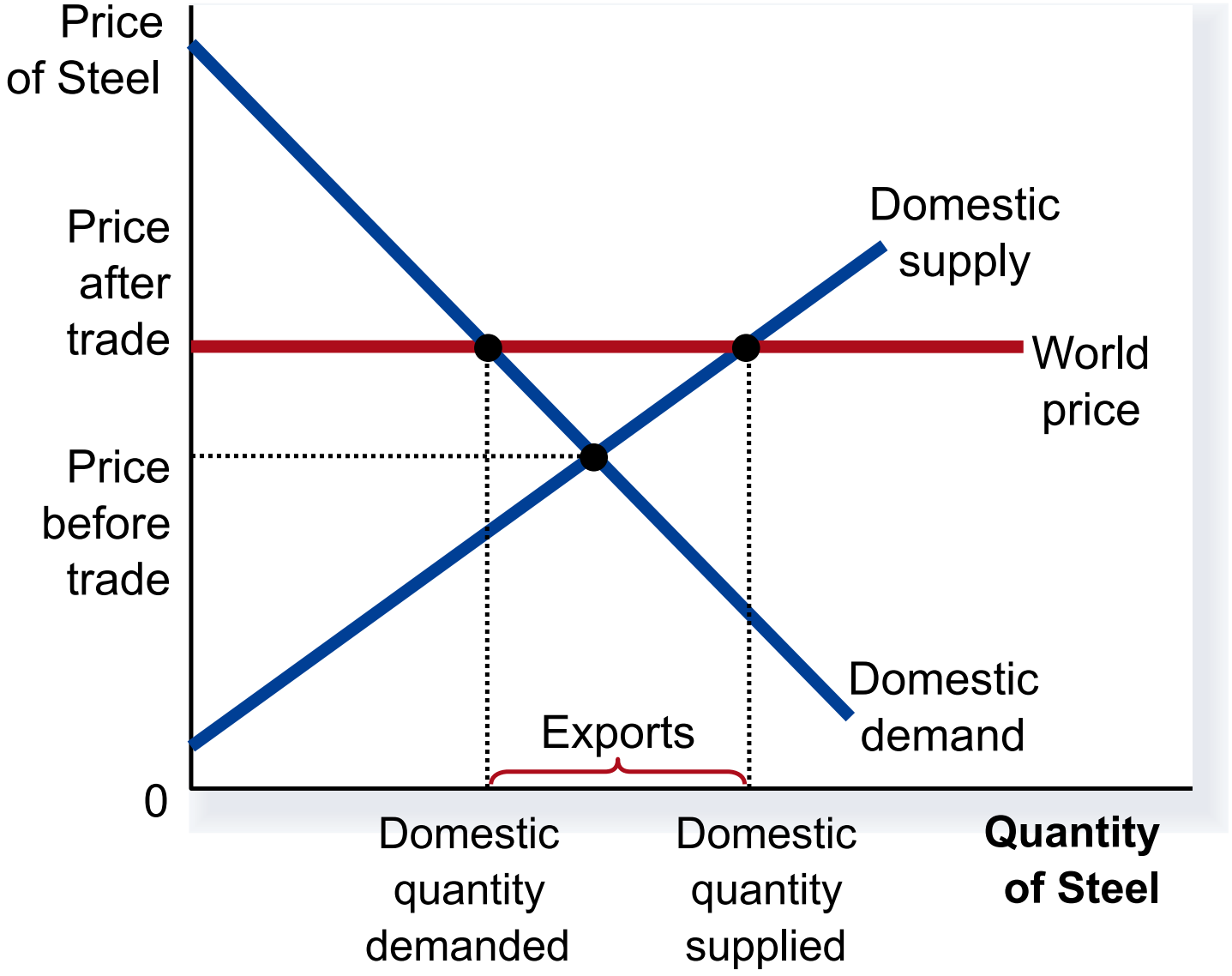
The Equilibrium without International Trade



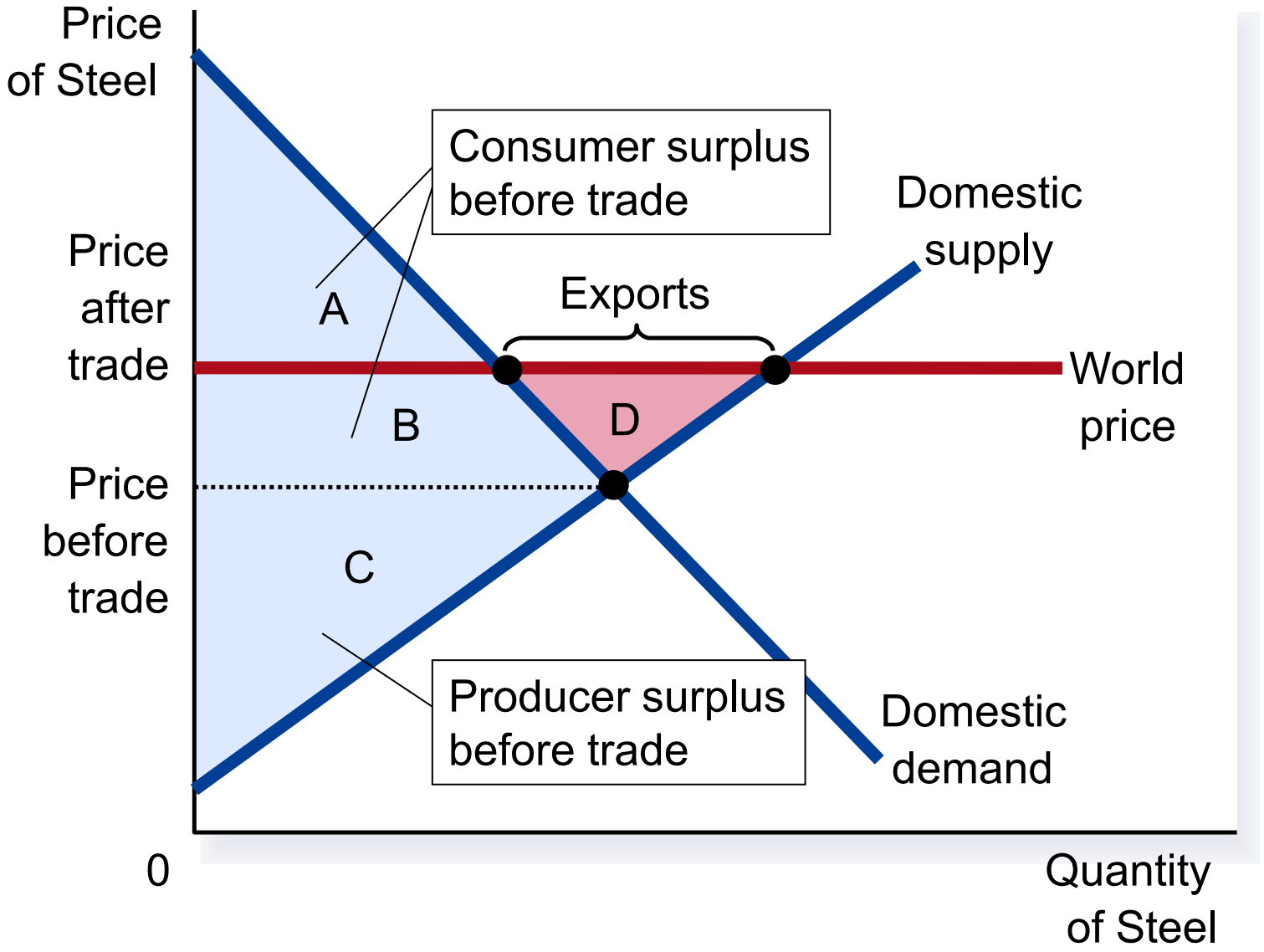
The World Price and Comparative Advantage

- The effects of free trade can be shown by comparing the domestic price of a good without trade and the *world price* of the good
  - The world price refers to the price that prevails in the world market for that good
  - It reflects the value of the good relative to all other goods in the world market
- If a country has a comparative advantage, then the domestic price will be below the world price, and the country will be an ***exporter*** of the good
- If the country does not have a comparative advantage, then the domestic price will be higher than the world price, and the country will be an ***importer*** of the good

International Trade in an Exporting Country



How Free Trade Affects Welfare in an Exporting Country



How Free Trade Affects Welfare in an Exporting Country

	Before Trade	After Trade	Change
Consumer Surplus	A + B	A	−B
Producer Surplus	C	B + C + D	+(B + D)
Total Surplus	A + B + C	A + B + C + D	+D

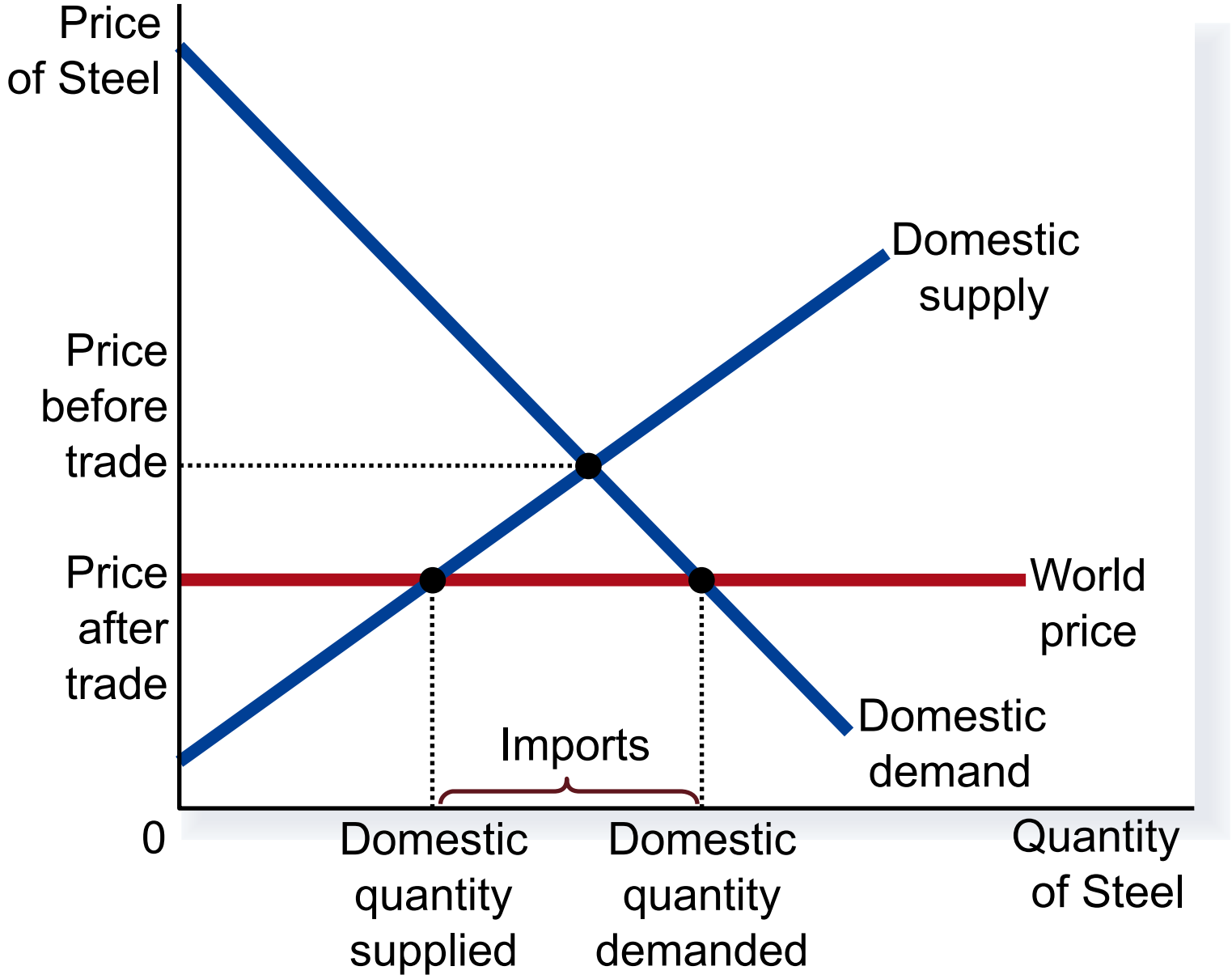
The area D shows the increase in total surplus and represents the gains from trade.

- Domestic producers of the good are better off, and domestic consumers of the good are worse off
- Trade raises the economic well-being of the nation as a whole

The Gains and Losses of an Importing Country

- International Trade in an Importing Country
  - If the world price of steel is lower than the domestic price, the country will be an importer of steel when trade is permitted
  - Domestic consumers will want to buy steel at the lower world price
  - Domestic producers of steel will have to lower their output because the domestic price moves to the world price

International Trade in an Importing Country



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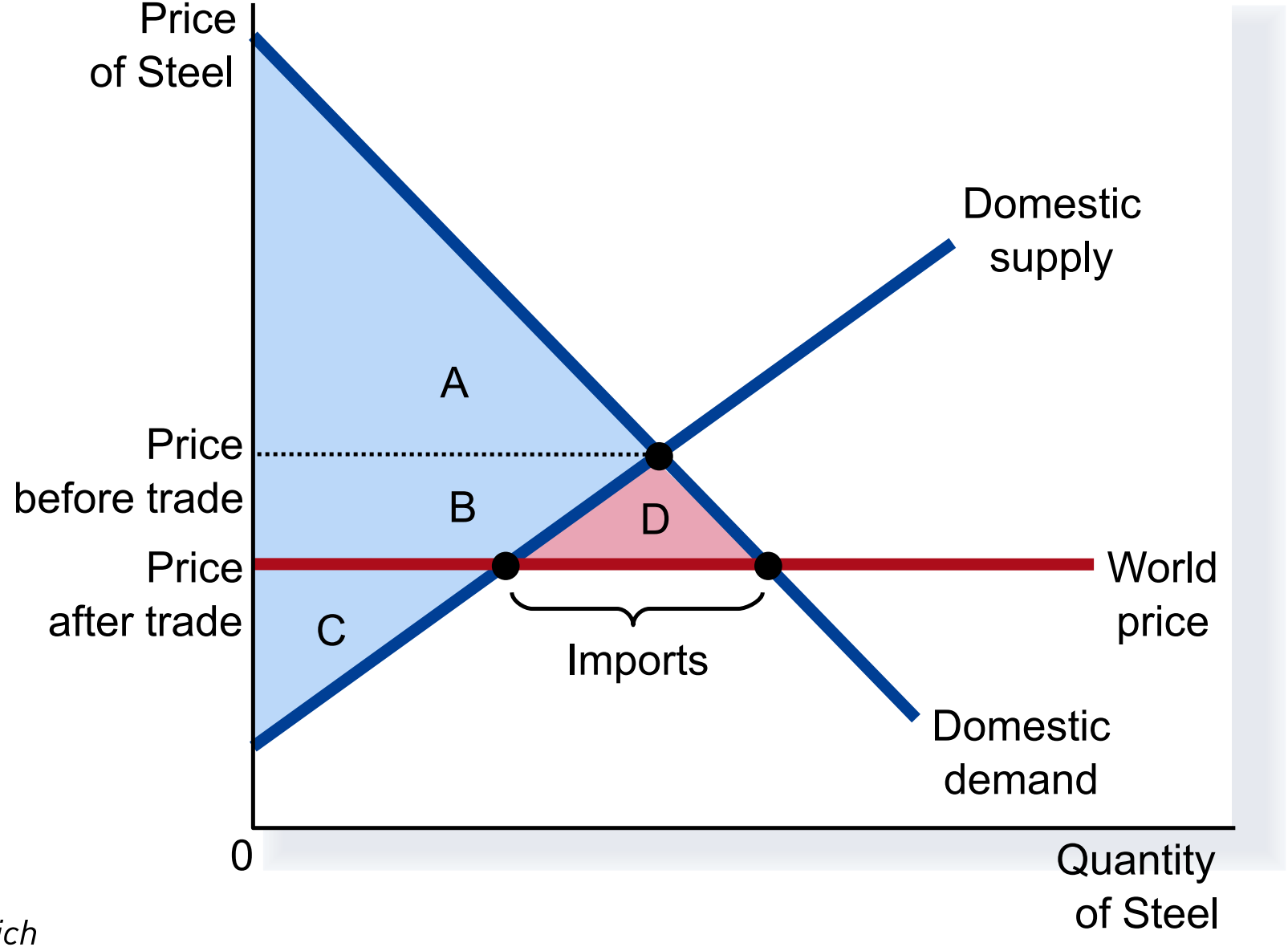

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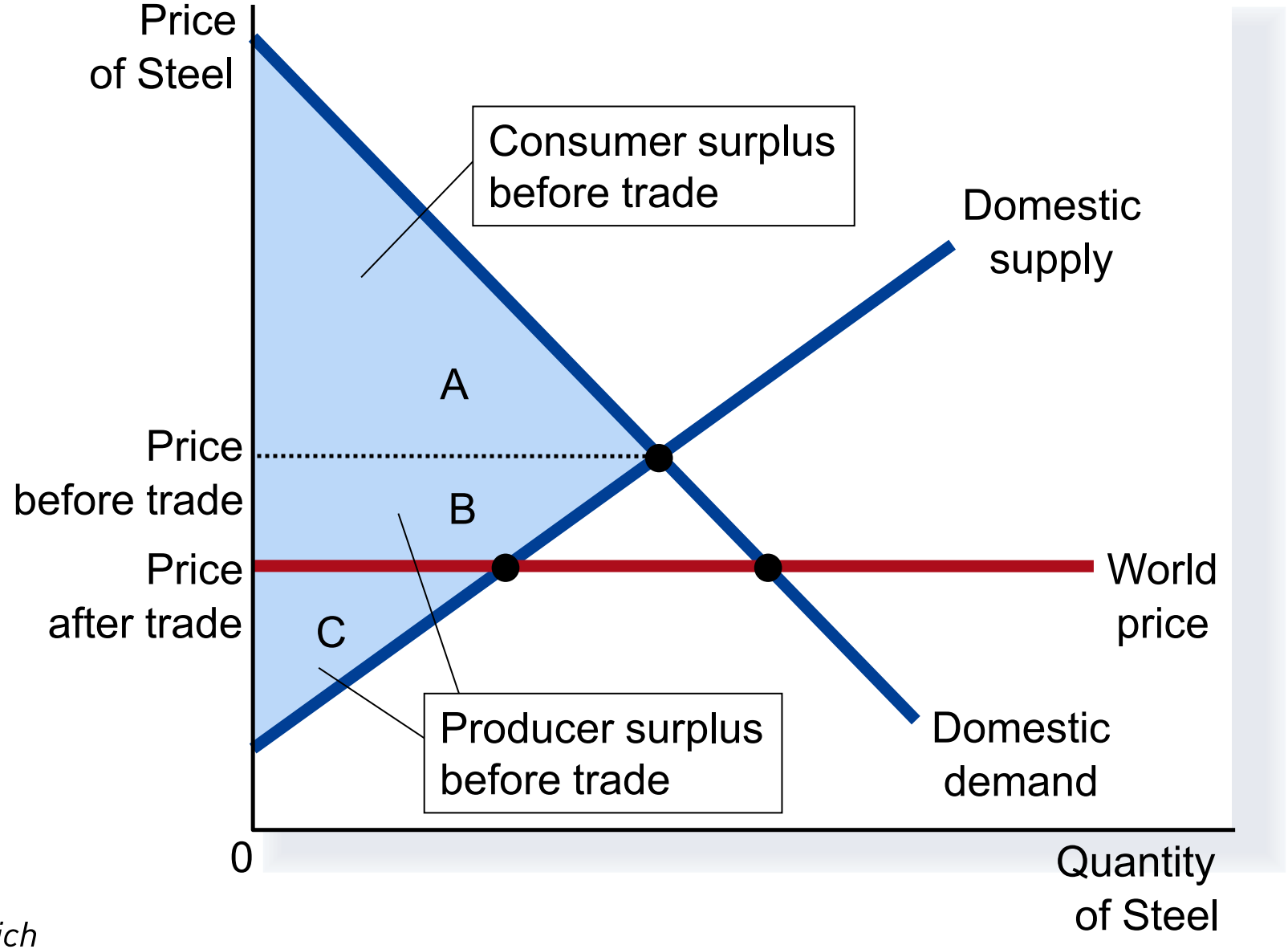
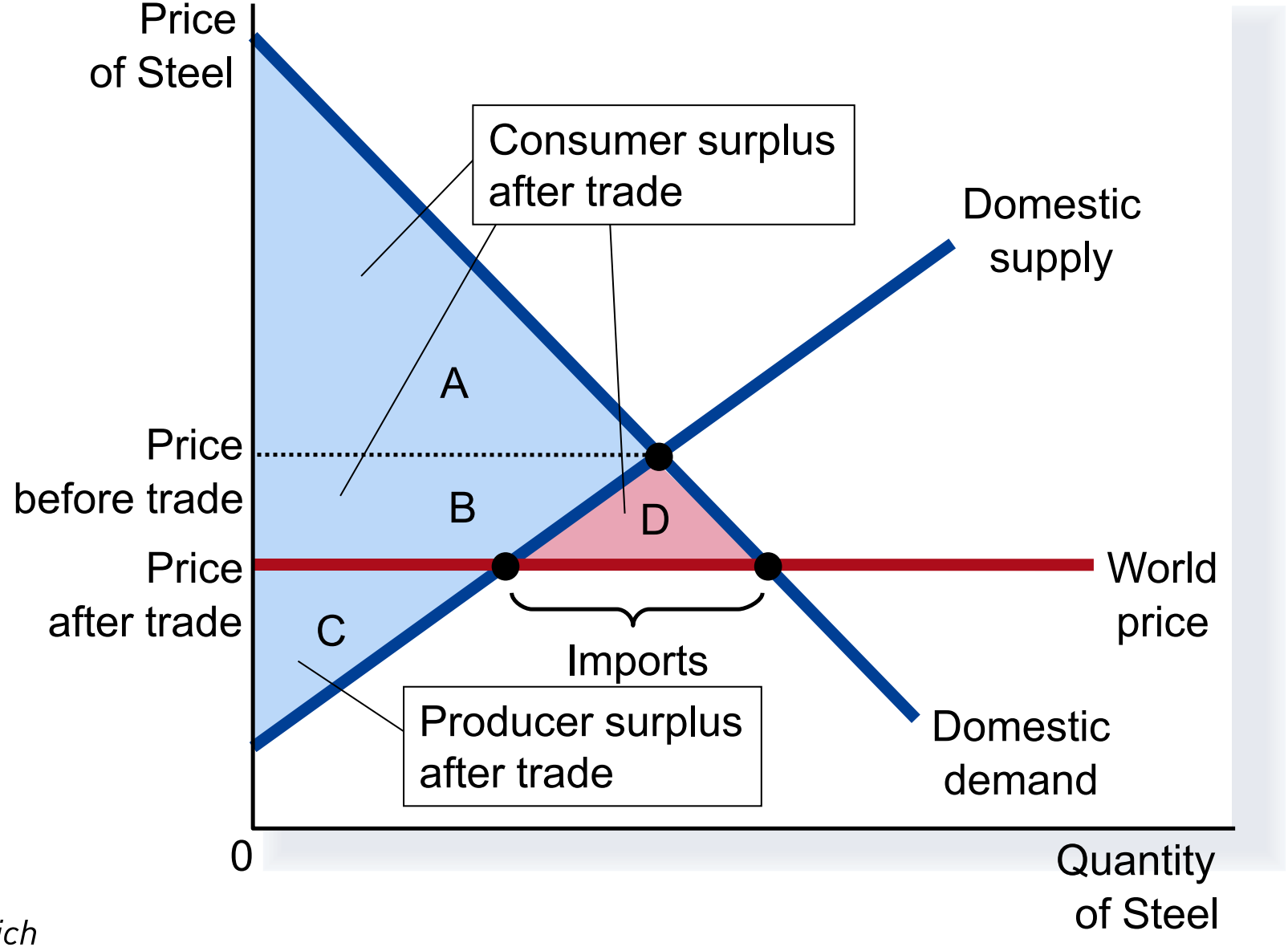


How Free Trade Affects Welfare in an Importing Country



How Free Trade Affects Welfare in an Importing Country

	Before Trade	After Trade	Change
Consumer Surplus	A	A + B + D	+(B + D)
Producer Surplus	B + C	C	−B
Total Surplus	A + B + C	A + B + C + D	+D

The area D shows the increase in total surplus and represents the gains from trade.

- Domestic producers of the good are worse off, and domestic consumers of the good are better off
- Trade raises the economic well-being of the nation as a whole because the gains of consumers exceed the losses of producers
- The gains of the winners exceed the losses of the losers
- The net change in total surplus is positive

FYI: Other benefits of International Trade

- Increased variety of goods
- Lower costs through economies of scale
- Increased competition
- Enhanced flow of ideas

Restrictions On Trade

Restrictions on trade come in the form of:

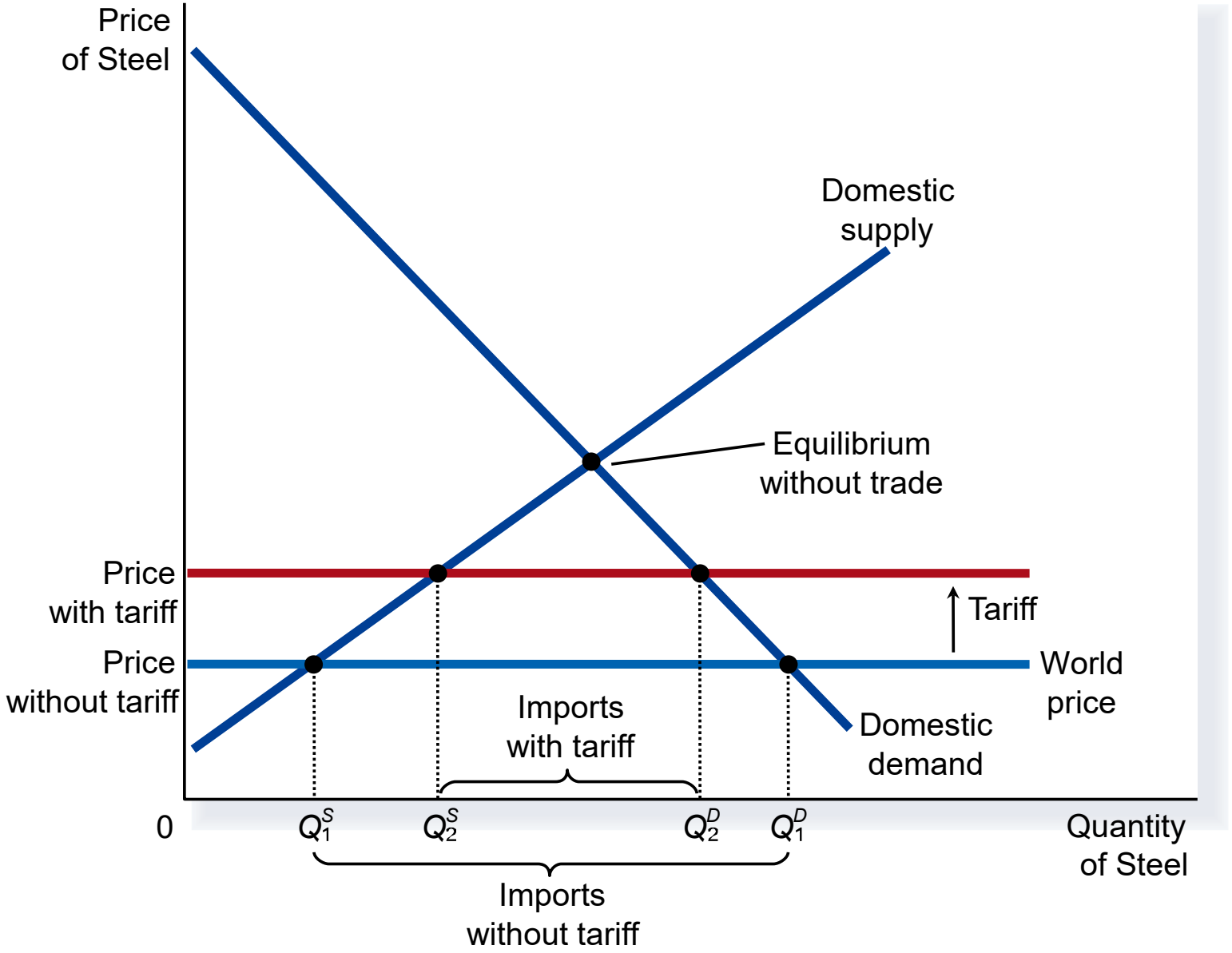
- Tariffs
- Quotas
- Non-tariff barriers

The Effects of a Tariff

- A ***tariff*** is a tax on goods produced abroad and sold domestically.
- Tariffs raise the price of imported goods above the world price by the amount of the tariff.

- Interested in “Trumpian Tariffs”? Theoretical extensions can be found here:
  - https://www.linkedin.com/pulse/teaching-trumpian-tariffs-richard-baldwin-ksjce/

The Effects of a Tariff



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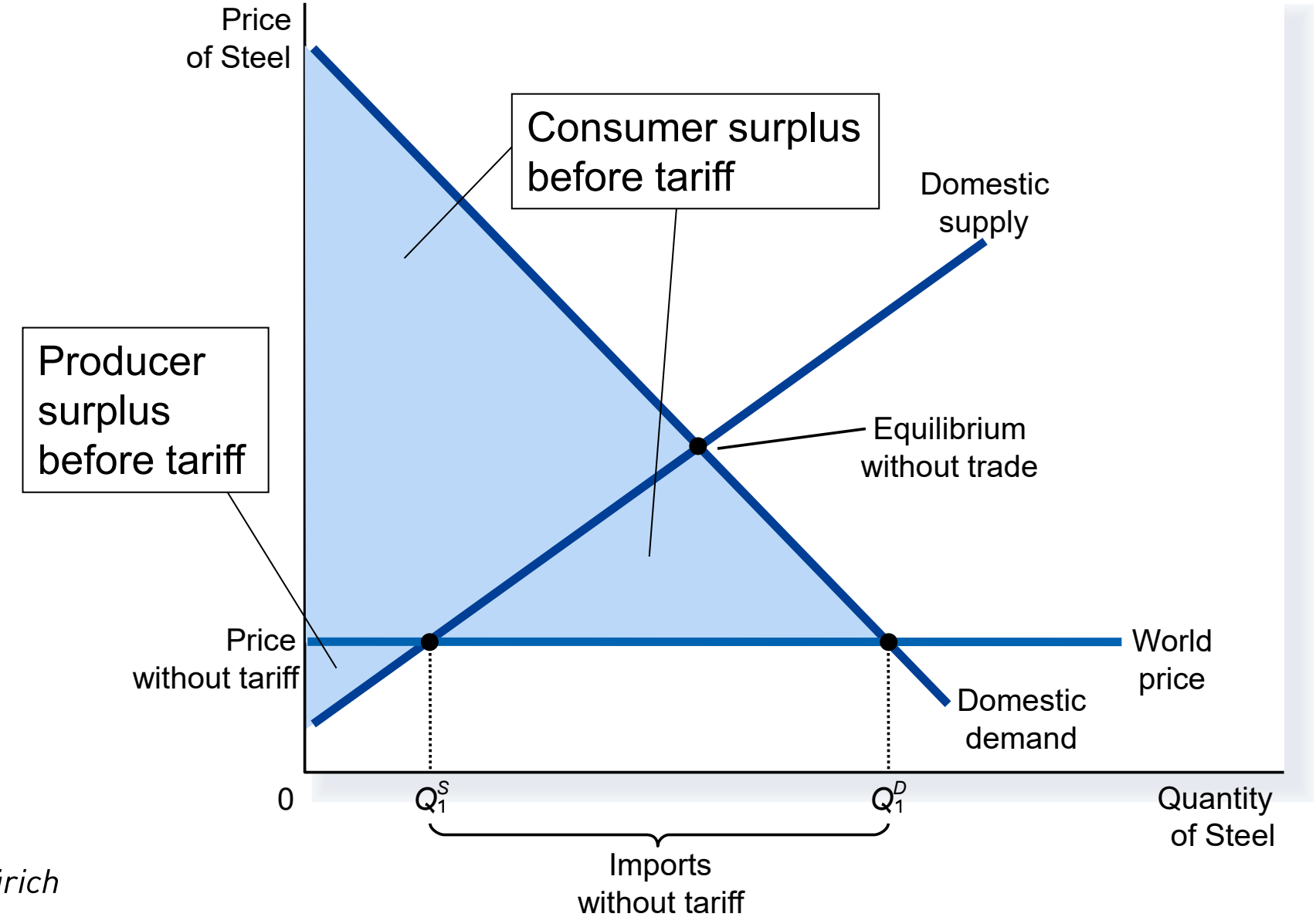

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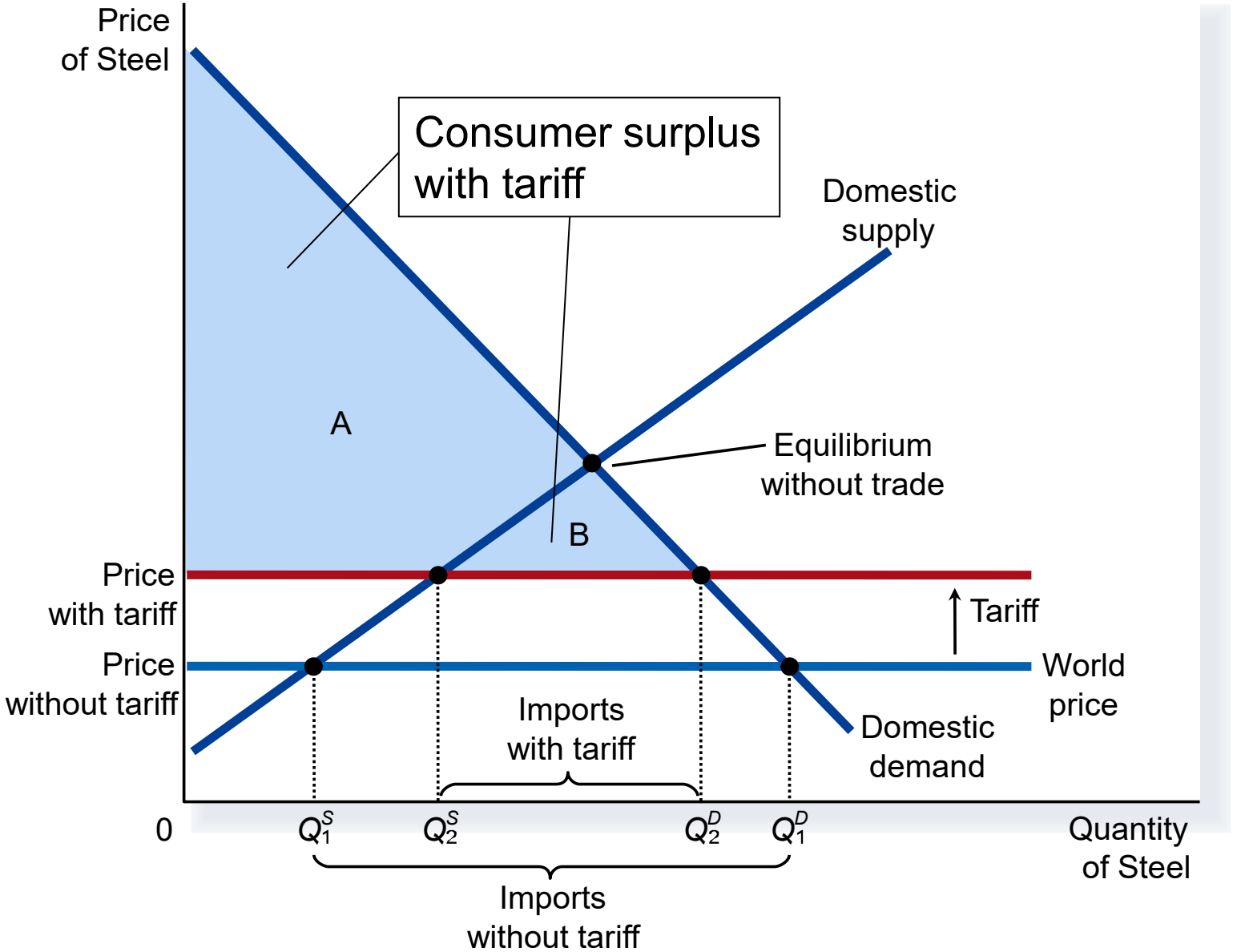

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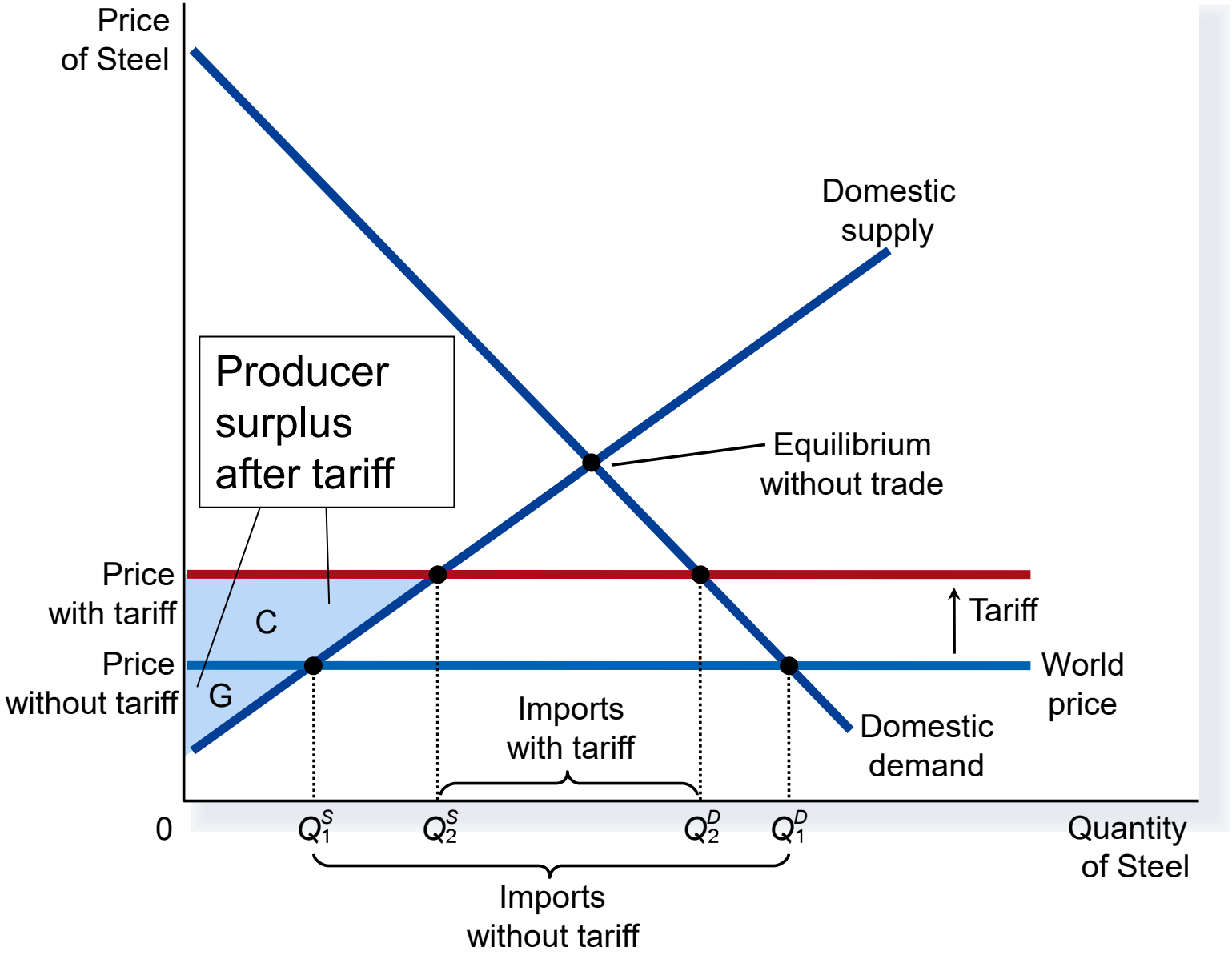
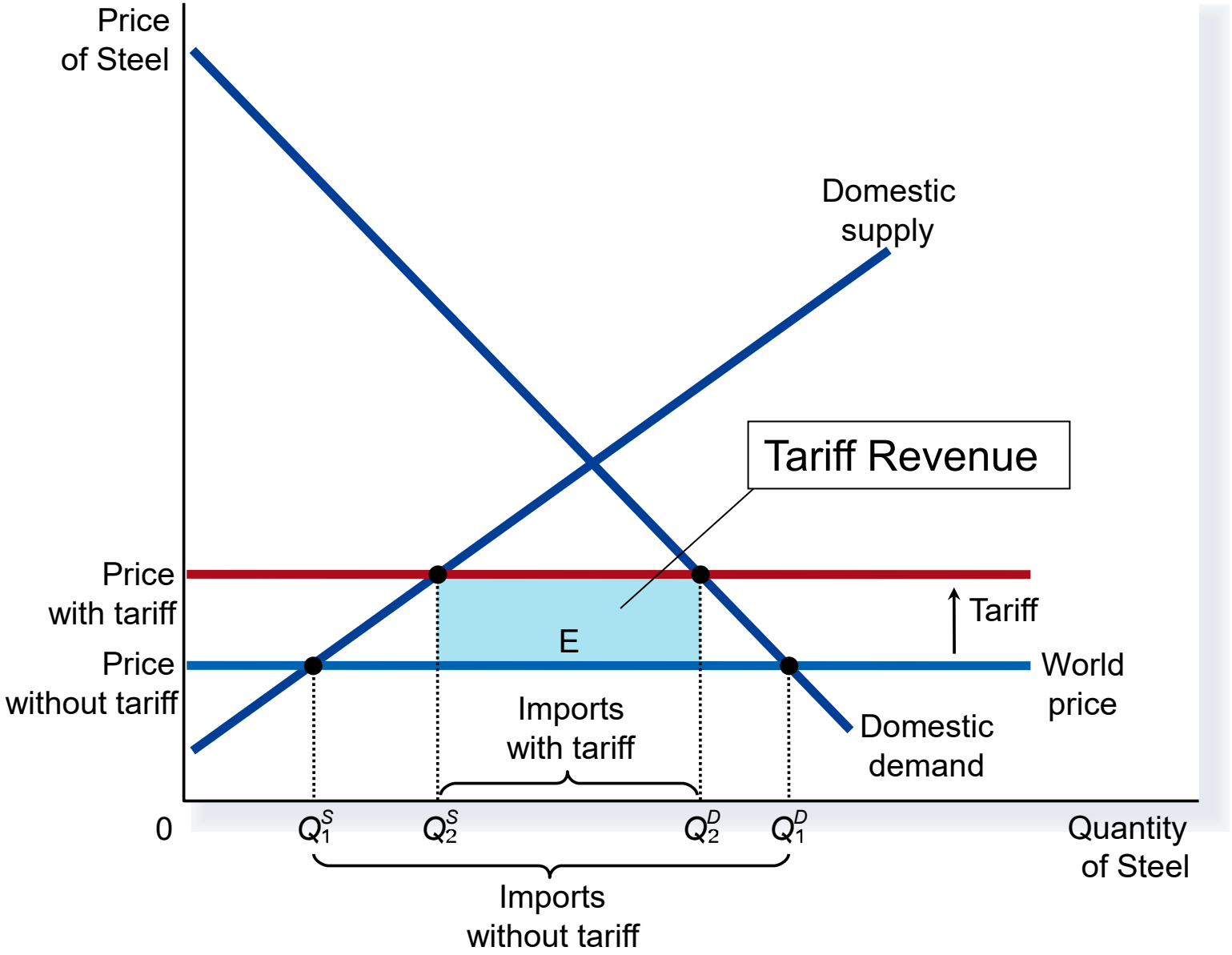


The Effects of a Tariff



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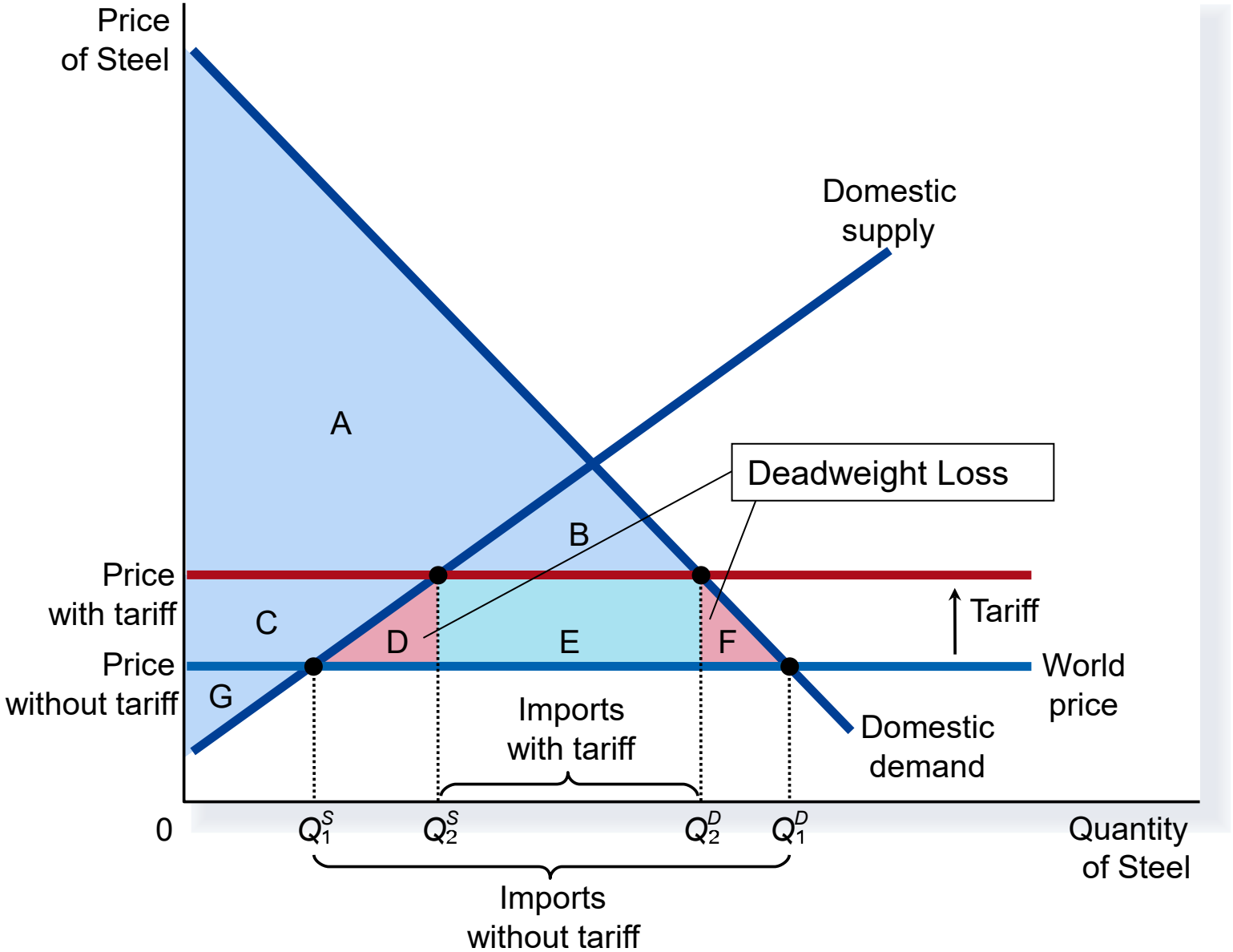


The Effects of a Tariff

	Before Tariff	After Tariff	Change
Consumer Surplus	A + B + C + D + E + F	A + B	−(C + D + E + F)
Producer Surplus	G	C + G	+C
Government Revenue	None	E	+E
Total Surplus	A + B + C + D + E + F + G	A + B + C + E + G	−(D + F)

The area D + F shows the fall in total surplus and represents the deadweight loss of the tariff.

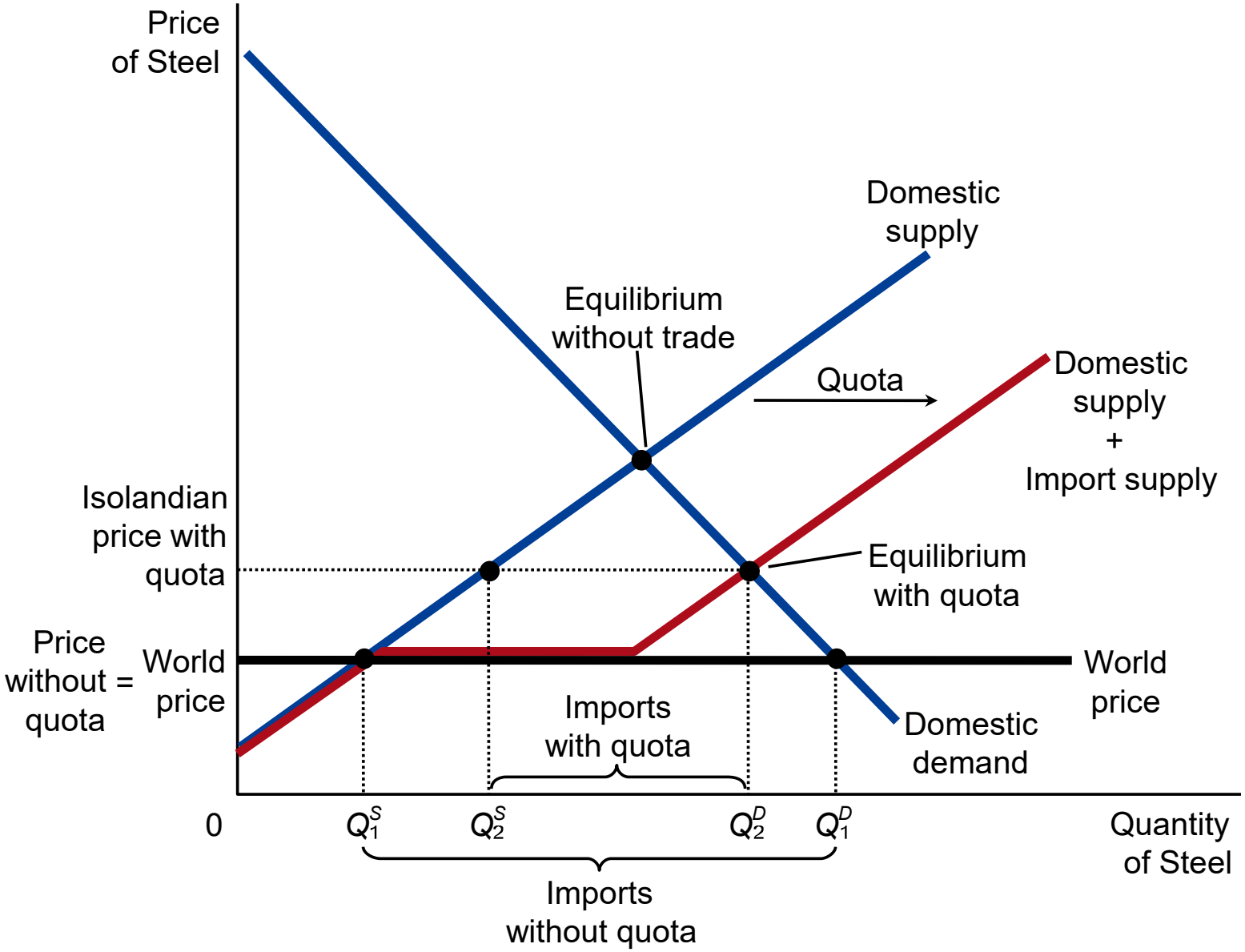
- A tariff reduces the quantity of imports and moves the domestic market closer to its equilibrium without trade
- With a tariff, total surplus in the market decreases by an amount referred to as a deadweight loss

The Effects of an Import Quota

- An ***import quota*** is a limit on the quantity of a good that can be produced abroad and sold domestically

- Because the quota raises the domestic price above the world price,
  domestic buyers of the good are worse off,
  and domestic sellers of the good are better off
- License holders are better off
  because they make a profit from buying at the world price
  and selling at the higher domestic price

The Effects of an Import Quota



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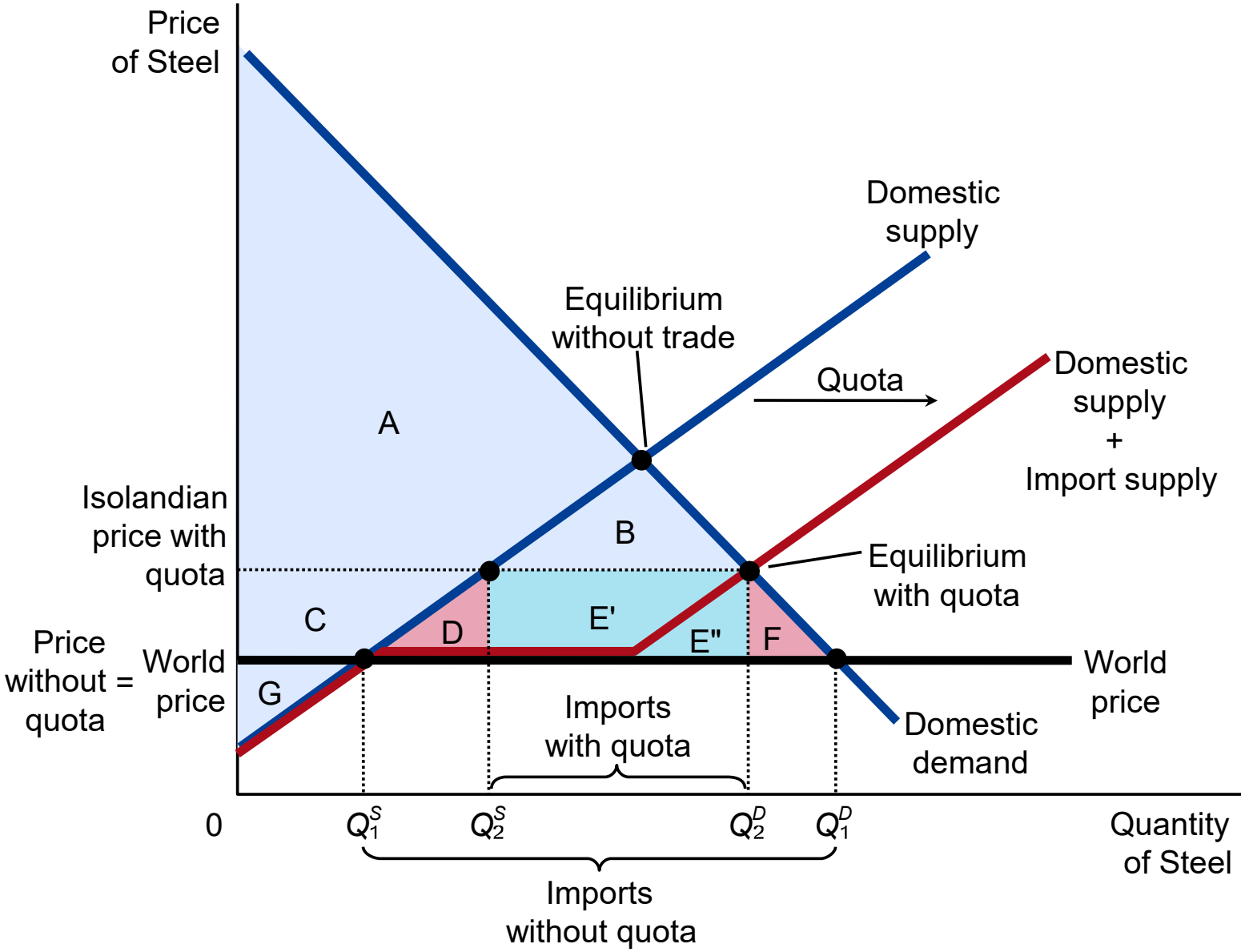


The Effects of an Import Quota

	Before Quota	After Quota	Change
Consumer Surplus	A + B + C + D + E' + E'' + F	A + B	−(C + D + E' + E'' + F)
Producer Surplus	G	C + G	+C
License-Holder Surplus	None	E' + E''	+(E' + E'')
Total Surplus	A + B + C + D + E' + E'' + F + G	A + B + C + E' + E'' + G	−(D + F)

The area D + F shows the fall in total surplus and represents the deadweight loss of the quota.

- With a quota, total surplus in the market decreases by an amount referred to as a deadweight loss
- The quota can potentially cause an even larger deadweight loss, if a mechanism such as lobbying is employed to allocate the import licenses

The Lessons for Trade Policy

- If government sells import licences for full value, revenue equals that of an equivalent tariff and the results of tariffs and quotas are identical

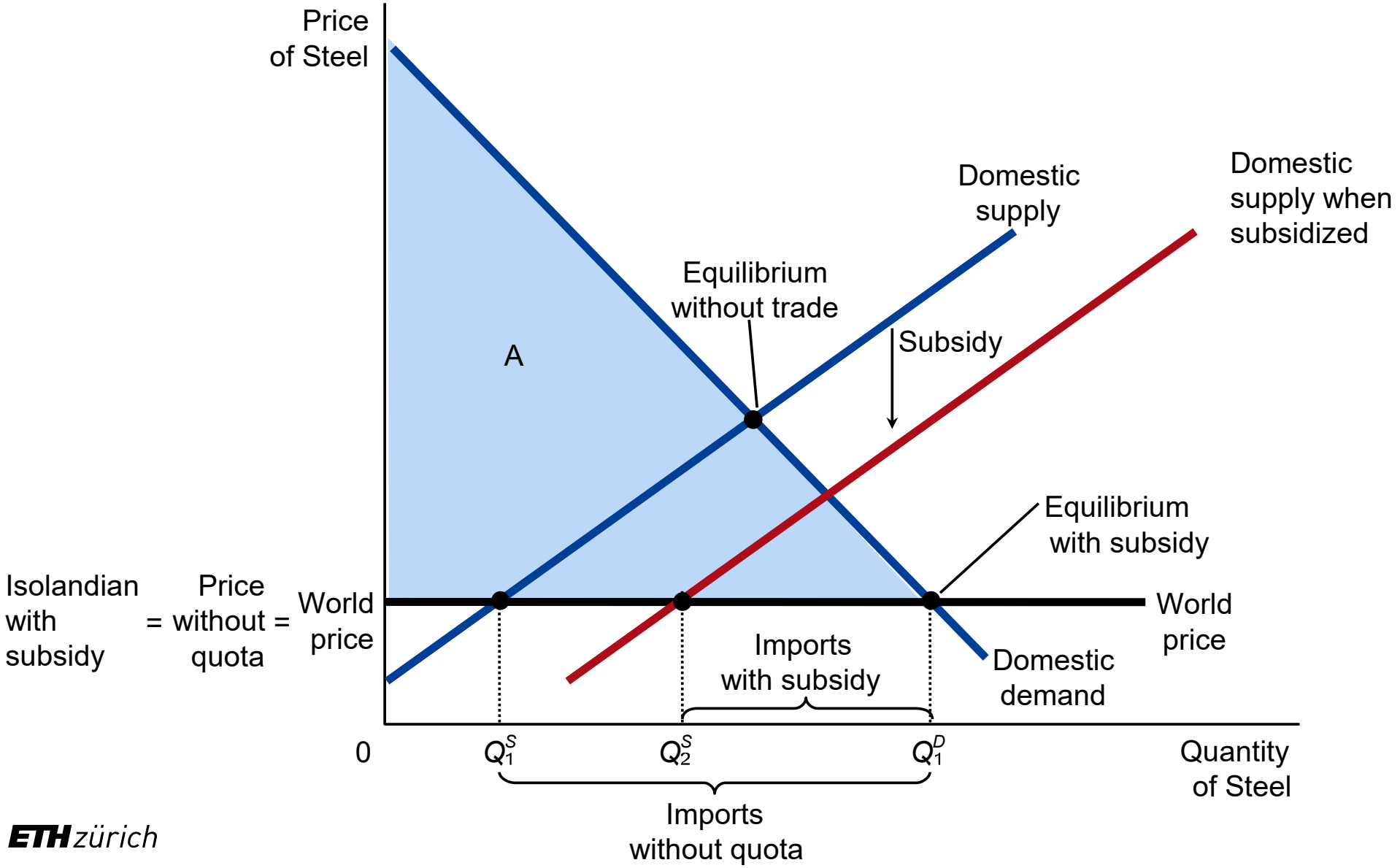
- Both tariffs and import quotas . . .
  - raise domestic prices
  - reduce the welfare of domestic consumers
  - increase the welfare of domestic producers
  - cause deadweight losses

- Are there other ways to support domestic producers?

The Effects of a Domestic Producer Subsidy

- An ***subsidy*** reduces the market price that a firm needs to receive in order to be willing to produce a good

The Effects of a Domestic Producer Subsidy



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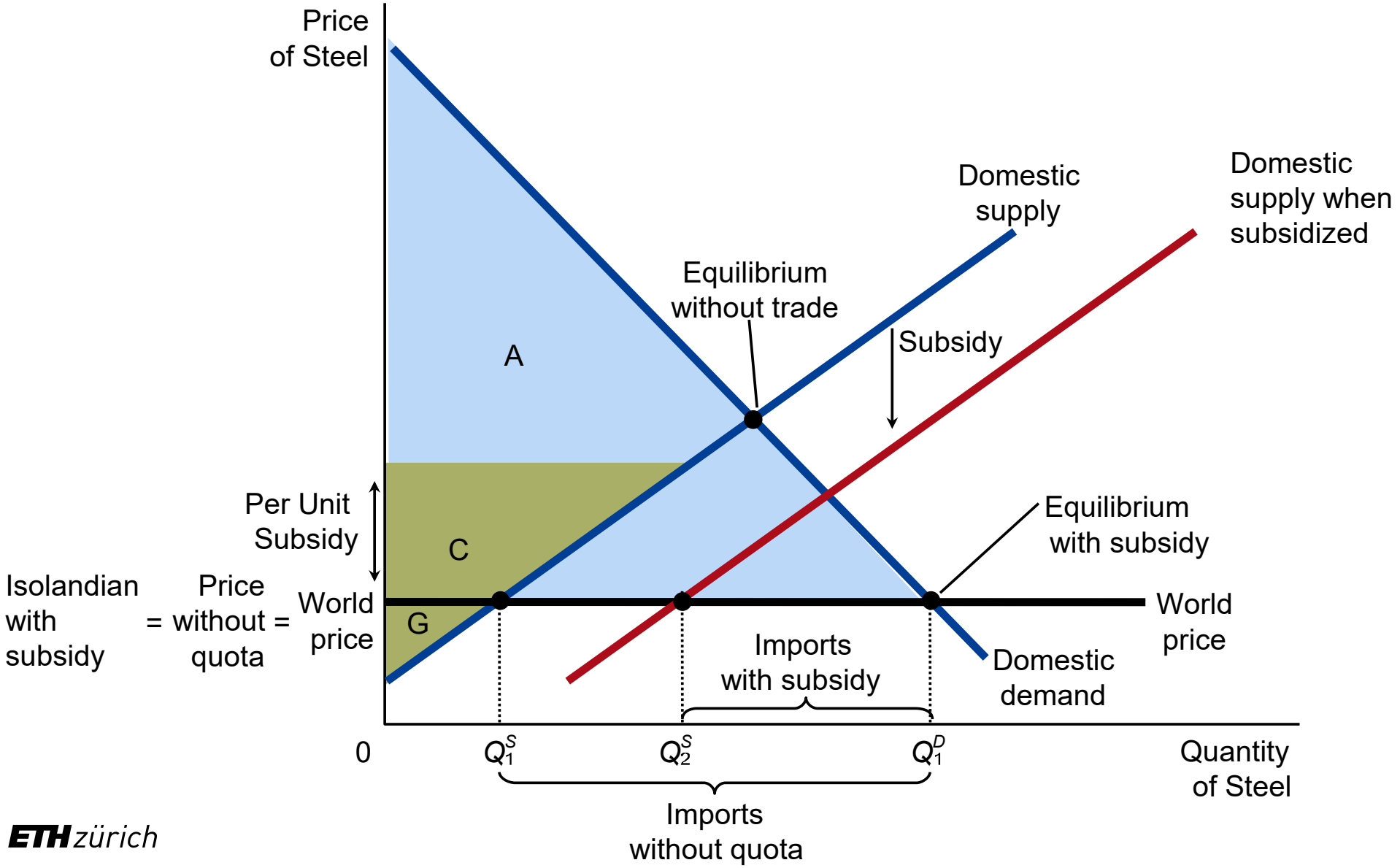

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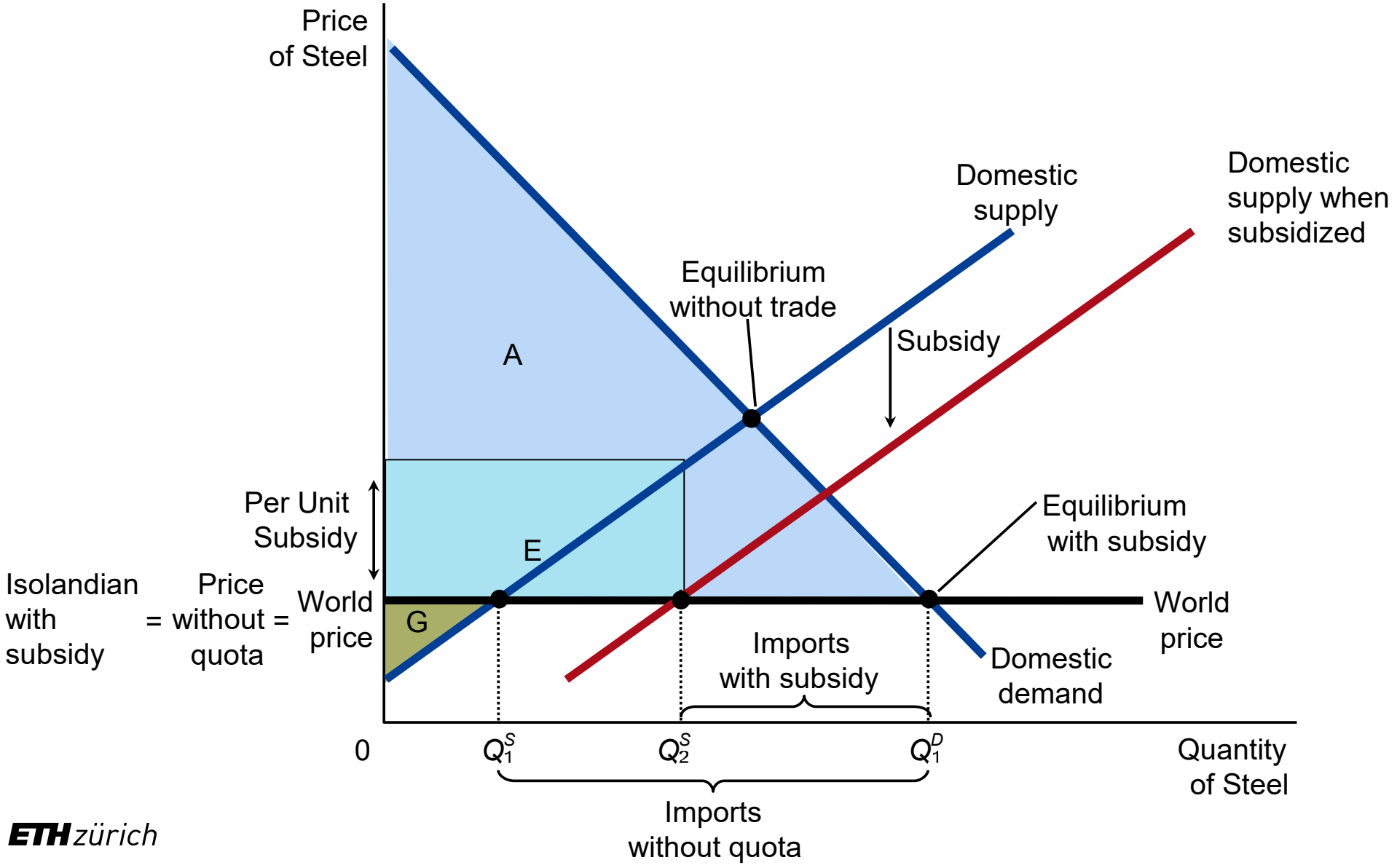

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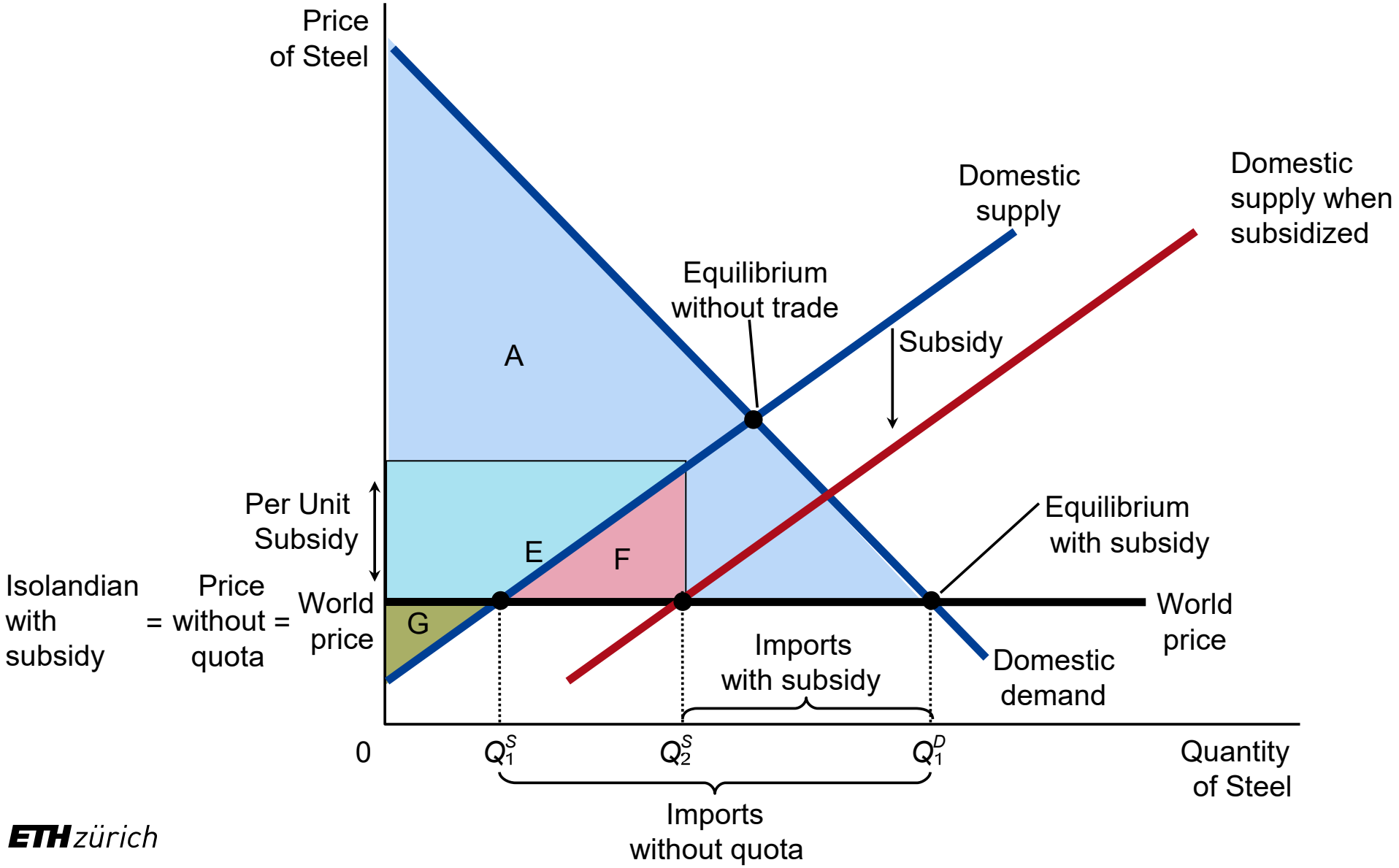

The Effects of Domestic Producer Subsidy

	Before Domestic Producer Subsidy	After Domestic Producer Subsidy	Change
Consumer Surplus	A	A	0
Producer Surplus	G	G+C	+C
Government Revenue/Costs	None	–E	–E
Change	A+G	A+G+C-E	C-E (= –F)

- A producer subsidy reduces the quantity of imports, while keeping the domestic price equal to the world price
- With such a subsidy, total surplus in the market decreases by an amount referred to as a deadweight loss

Non-Tariff Barriers

- Complex or discriminatory rules of origin and quality conditions
  - Technical and health and safety issues
- Sanitary and phytosanitary measures
  - Measures to protect humans, animals, and plants from diseases, pests, or contaminants, like food safety standards
- Administrative regulations
  - Red tape
- Currency manipulation

The Arguments for Restricting Trade

- Climate change mitigation
- Food security
- Geopolitical concerns (like Unfair Competition argument)
- National Security argument
- Public health concerns
- Resilience
- Strategic competitiveness (Infant Industry argument)
- Other (like Jobs argument, or Protection as a Bargaining Chip)

- See also: https://econofact.org/industrial-policy-is-back-is-that-a-good-thing

Summary

- The effects of free trade can be determined by comparing the domestic price without trade to the world price
  - A low domestic price indicates that the country has a comparative advantage in producing the good; the country will become an exporter
  - A high domestic price indicates that other countries have a comparative advantage in producing the good; the country will become an importer
- When a country allows trade and becomes an exporter of a good, producers of the good are better off, and consumers of the good are worse off
- When a country allows trade and becomes an importer of a good, consumers of the good are better off, and producers are worse off
- There are various arguments for restricting trade:
  - protecting jobs, defending national security, helping infant industries, preventing unfair competition, responding to foreign trade restrictions
- Economists believe that free trade is in the long run the better policy

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Principles of Macroeconomics

Lecture 8: Open-economy macroeconomics

Prof. Dr. Jan-Egbert Sturm

Open-Economy Macroeconomics: Basic Concepts

- Open and Closed Economies
  - A closed economy is one that does not interact with other economies in the world
    - There are no exports, no imports, and no capital flows
  - An open economy is one that interacts freely with other economies around the world

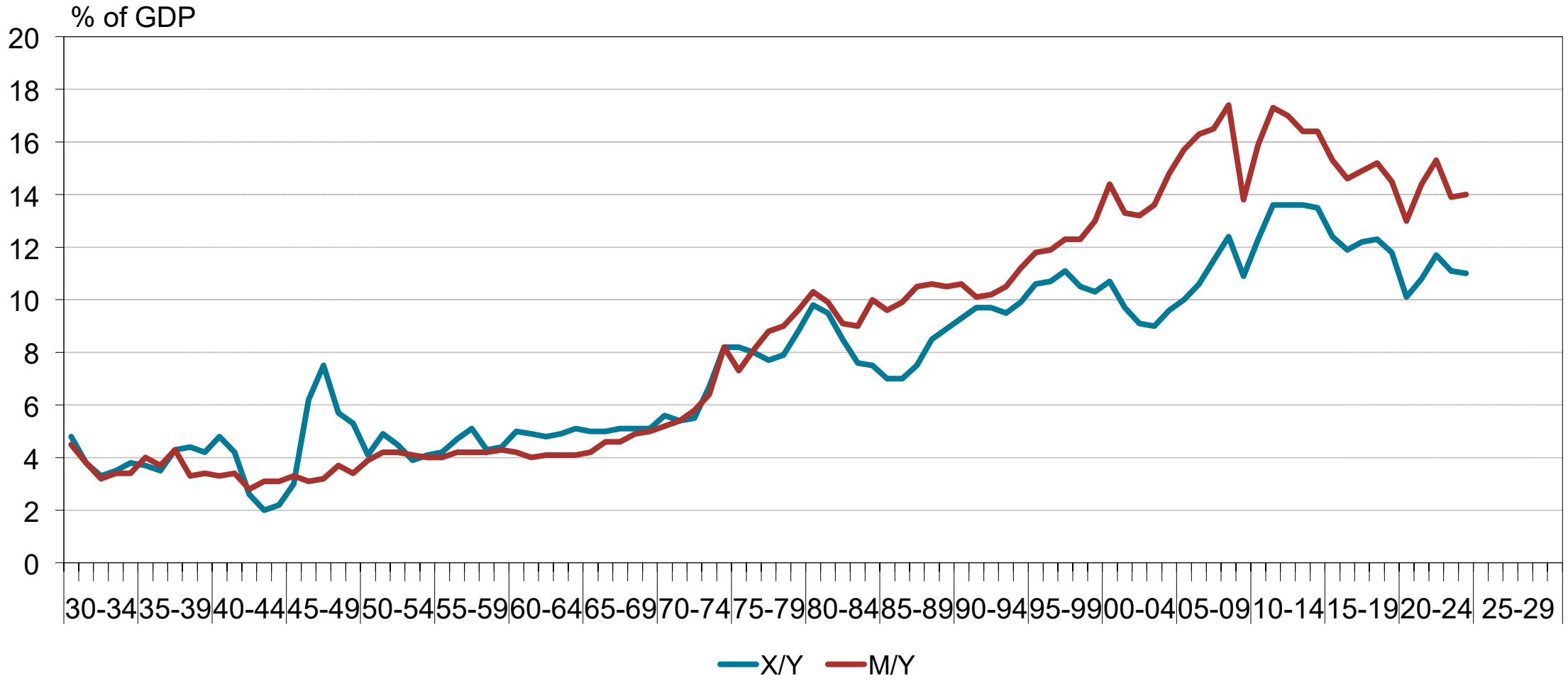
- An open economy interacts with other countries in two ways.
  - It buys and sells goods and services in world product markets
  - It buys and sells capital assets in world financial markets

- The Flow of Goods: Exports, Imports, and Net Exports
  - Switzerland is a small and relatively open economy
    — it imports and exports huge quantities of goods and services
  - Over the past four decades, international trade and finance have become increasingly important

The Flow of Goods: Exports, Imports, Net Exports

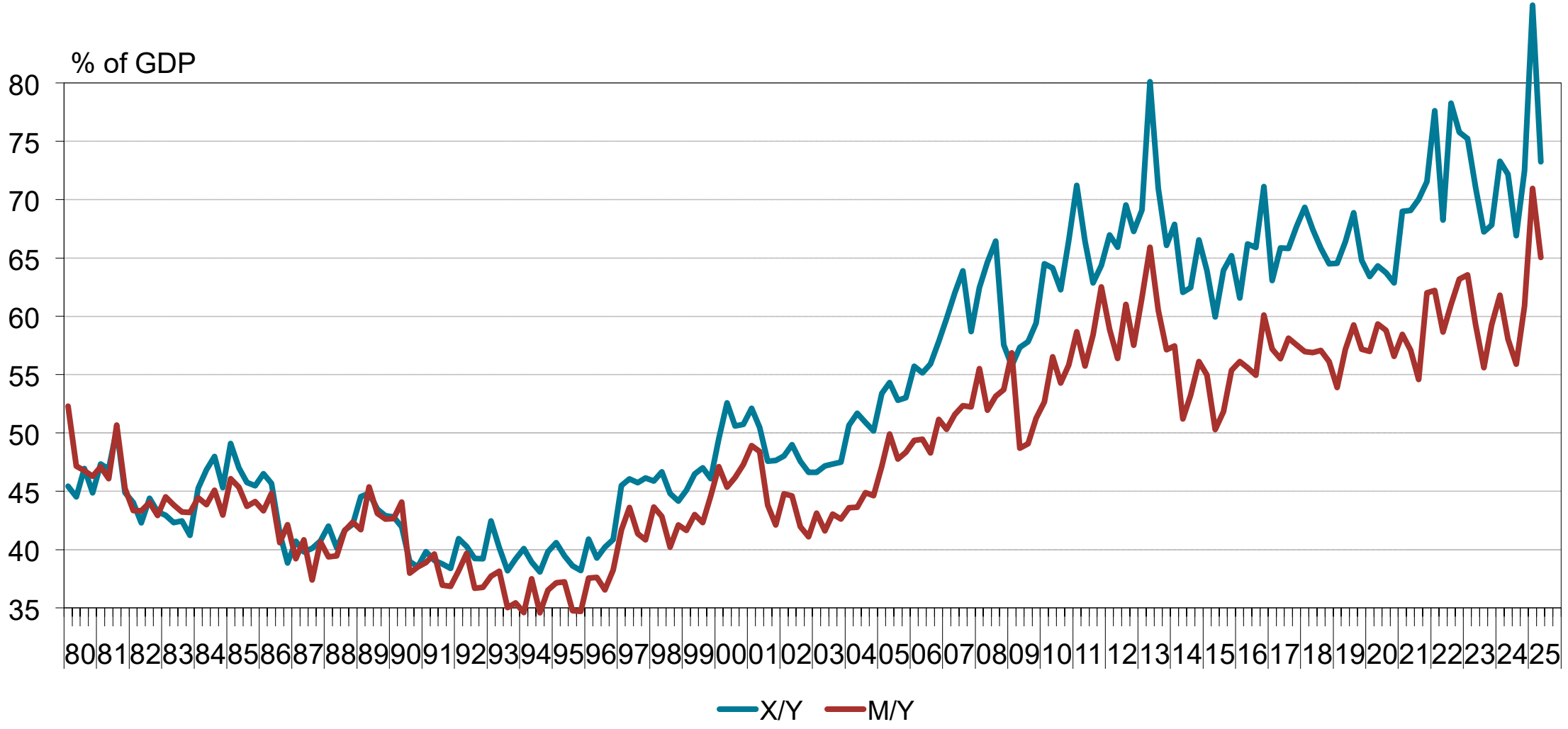
- ***Exports*** are goods and services that are produced domestically and sold abroad
- ***Imports*** are goods and services that are produced abroad and sold domestically
- ***Net exports*** (NX) are the value of a nation's exports minus the value of its imports
  - Net exports are also called the ***trade balance***
  - A trade (balance) deficit is a situation in which net exports (NX) are negative: Imports > Exports
  - A trade (balance) surplus is a situation in which net exports (NX) are positive: Exports > Imports
  - Balanced trade refers to when net exports are zero: Exports = Imports
  - Factors That Affect Net Exports
    - The tastes of consumers for domestic and foreign goods
    - The prices of goods at home and abroad
    - The exchange rates at which people can use domestic currency to buy foreign currencies
    - The incomes of consumers at home and abroad
    - The costs of transporting goods from country to country
    - The policies of the government toward international trade

The Internationalization of the U.S. Economy



Source: US BEA

The Internationalization of the Swiss Economy



Source: SECO

The Flow of Financial Resources: Net Capital Outflow

- ***Net capital outflow*** refers to the purchase of foreign assets by domestic residents minus the purchase of domestic assets by foreigners
  - A Swiss resident buys stock in the Cadbury's corporation and an American buys stock in Nestlé
    - When a Swiss resident buys stock in a British bank, the purchase raises Swiss net capital outflow
    - When an American resident buys a bond issued by the Swiss government, the purchase reduces Swiss net capital outflow

- Variables that Influence Net Capital Outflow
  - The real interest rates being paid on foreign assets
  - The real interest rates being paid on domestic assets
  - The exchange rate at which people can use domestic currency to buy foreign currencies
  - The perceived economic and political risks of holding assets abroad
  - The government policies that affect foreign ownership of domestic assets

The Equality of Net Exports and Net Capital Outflow

- For an economy as a whole, *NX* and *NCO* must balance each other so that:

$$NCO = NX$$

  - This holds true because every transaction that affects one side must also affect the other side by the same amount
- Net exports is a component of GDP:

$$Y = C + I + G + NX$$

- National saving is the income of the nation that is left after paying for current consumption and government purchases:

$$Y - C - G = I + NX$$

- National saving (S) equals $Y - C - G$ so:

$$S = I + NX$$
$$S - I = NX$$
$$S - I = \text{Net Capital Outflow} = NCO = NX$$

The Balance of Payments (BoP) Accounts

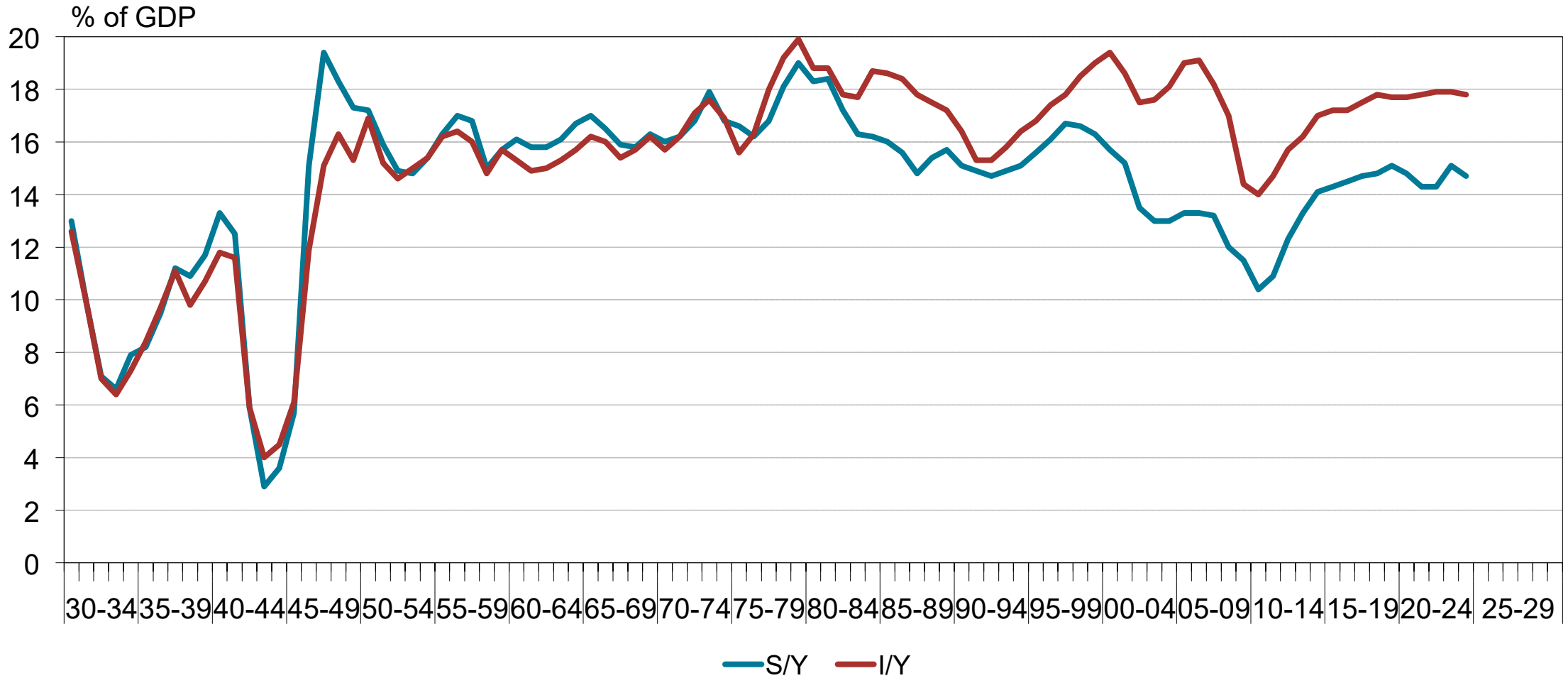
- The BoP consists of the current account (≈ trade account) and the financial (and capital) account
  - The sum of these accounts always equals zero
- Example of Paired Transactions
  - A Swiss firm buys a machine for CHF 1000 from a Danish firm
    - The Swiss firm transfers CHF 1000 to the Danish firm's bank account in Zurich
  - This transaction creates the following two offsetting entries in the Swiss balance of payments:
    - It enters the Swiss current account (NX) with a negative sign (ΔNX = - CHF 1000)
    - It shows up as a credit in the Swiss financial account (ΔNCO = - CHF 1000)
  - That is, Switzerland gets goods in return for assets

International Flows of Goods and Capital: Summary

This table shows the three possible outcomes for an open economy.

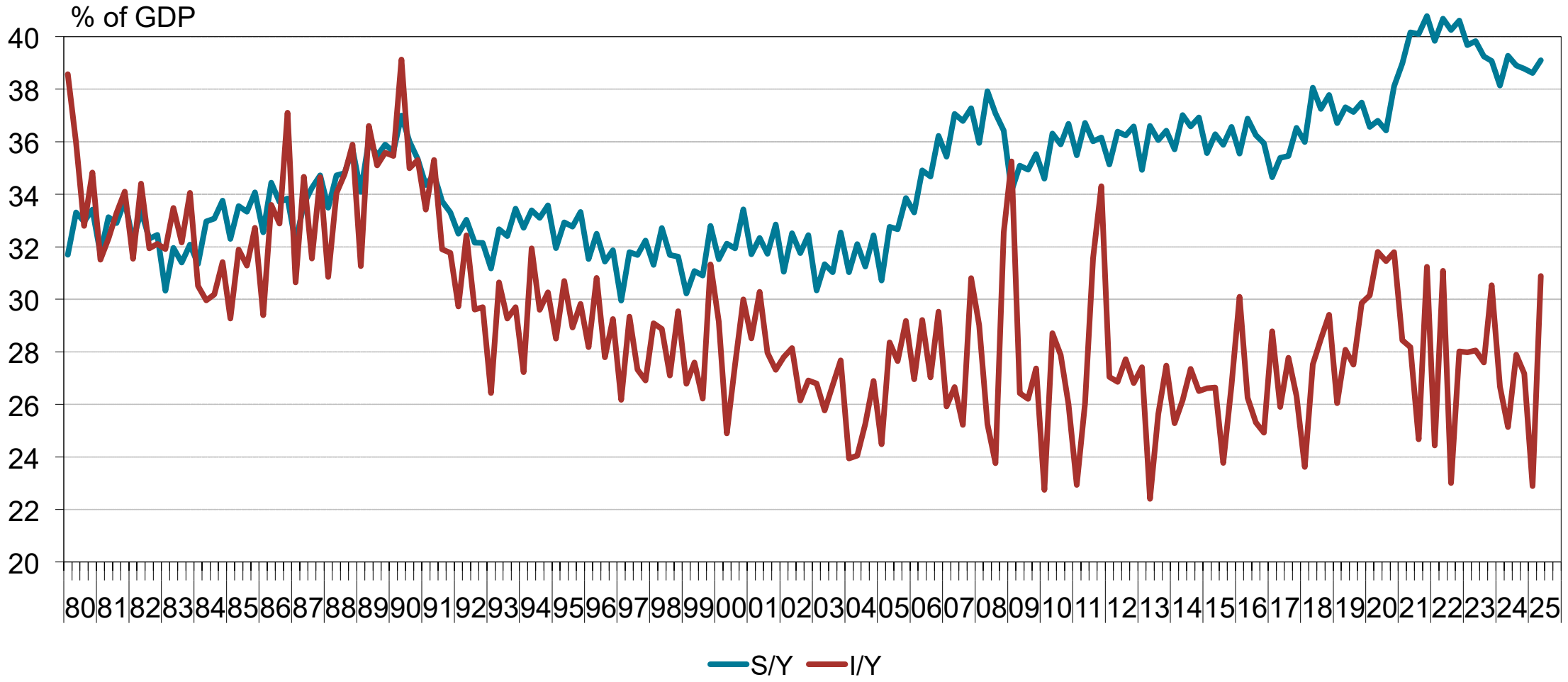
Trade Deficit	Balanced Trade	Trade Surplus
Exports < Imports	Exports = Imports	Exports > Imports
Net Exports < 0	Net Exports = 0	Net Exports > 0
$Y < C + I + G$	$Y = C + I + G$	$Y > C + I + G$
Saving < Investment	Saving = Investment	Saving > Investment
Net Capital Outflow < 0	Net Capital Outflow = 0	Net Capital Outflow > 0

National Saving and Domestic Investment in the US



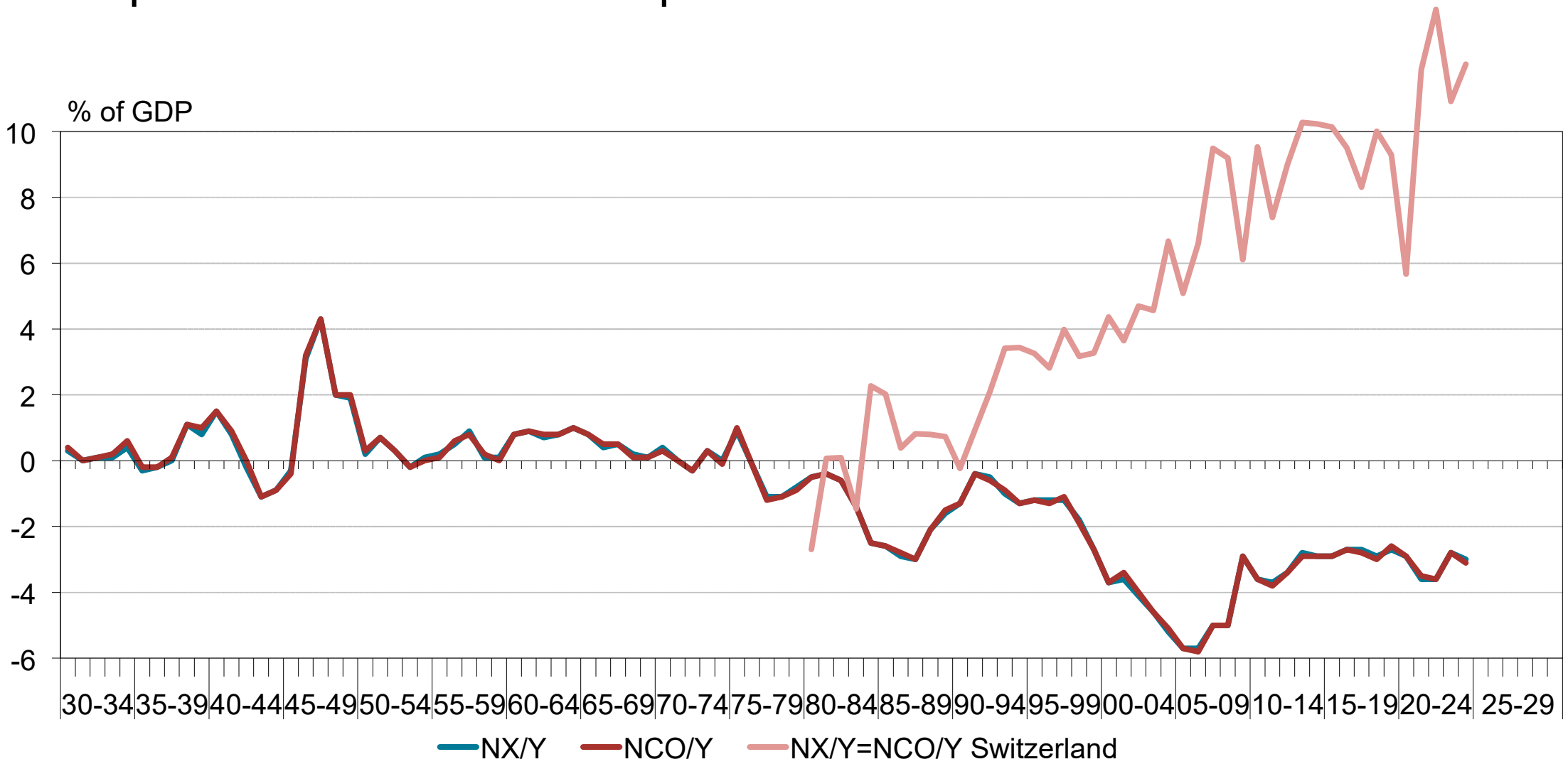
Source: US BEA

National Saving and Domestic Investment in Switzerland



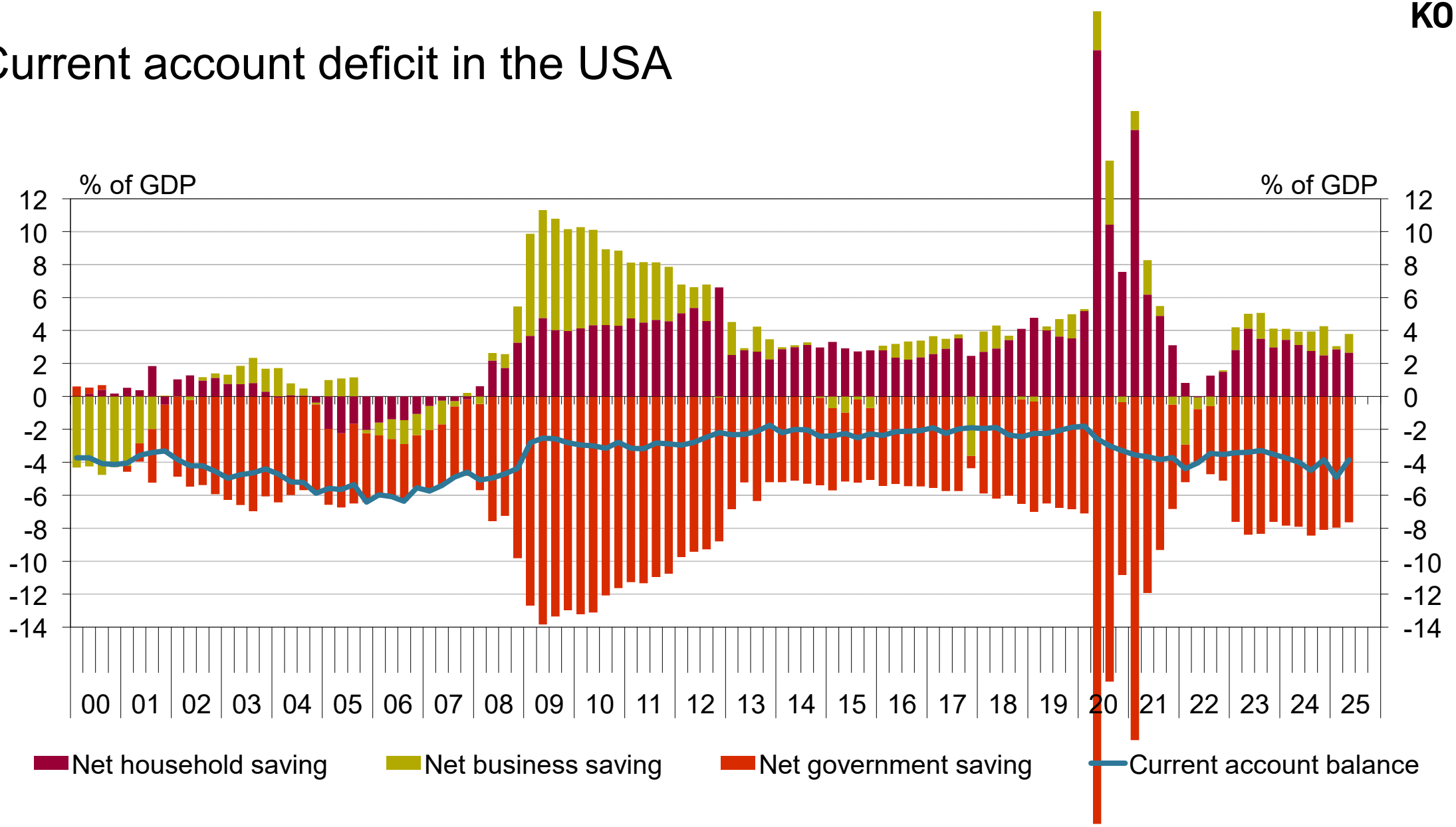
Source: SECO

Net Capital Outflow and Net Exports



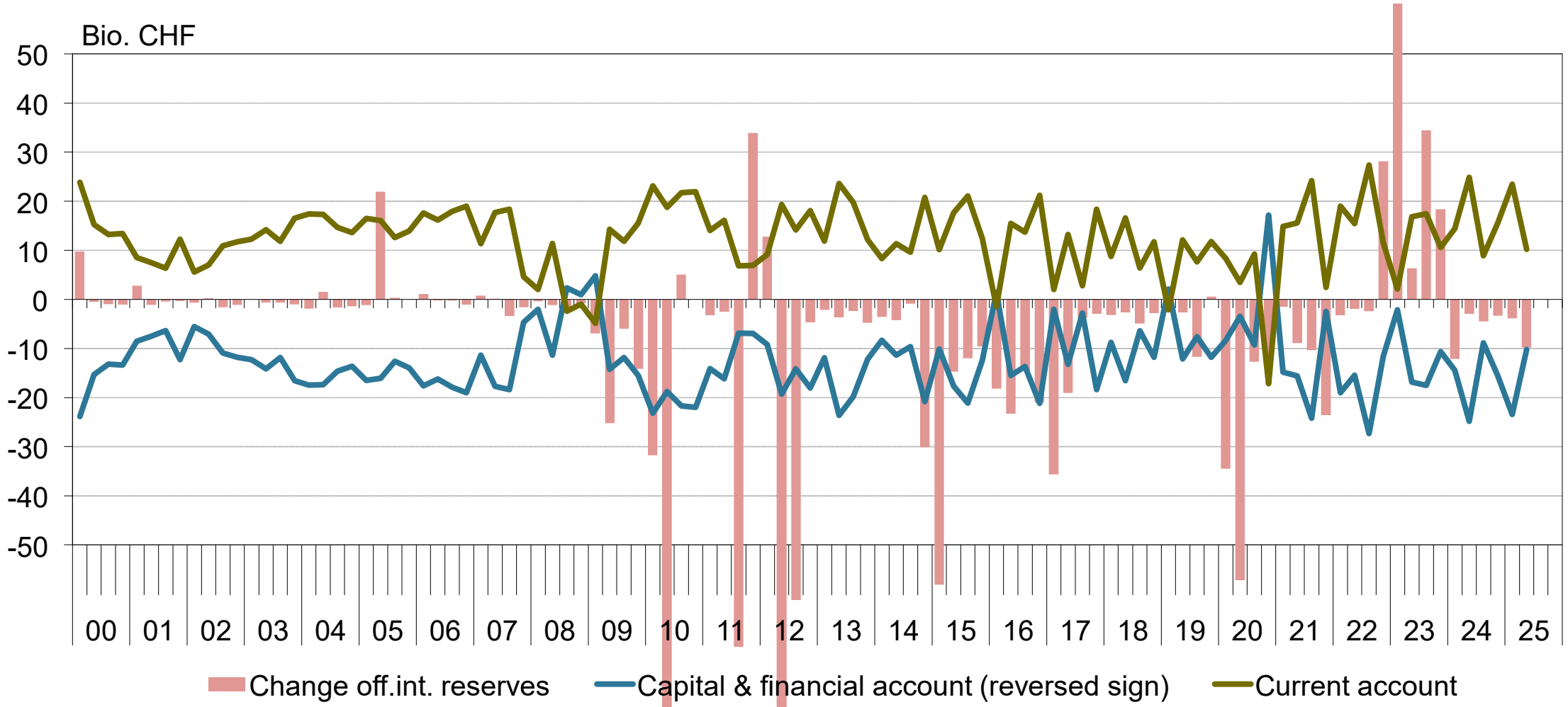
Source: US BEA, SECO

Current account deficit in the USA



Source: BEA

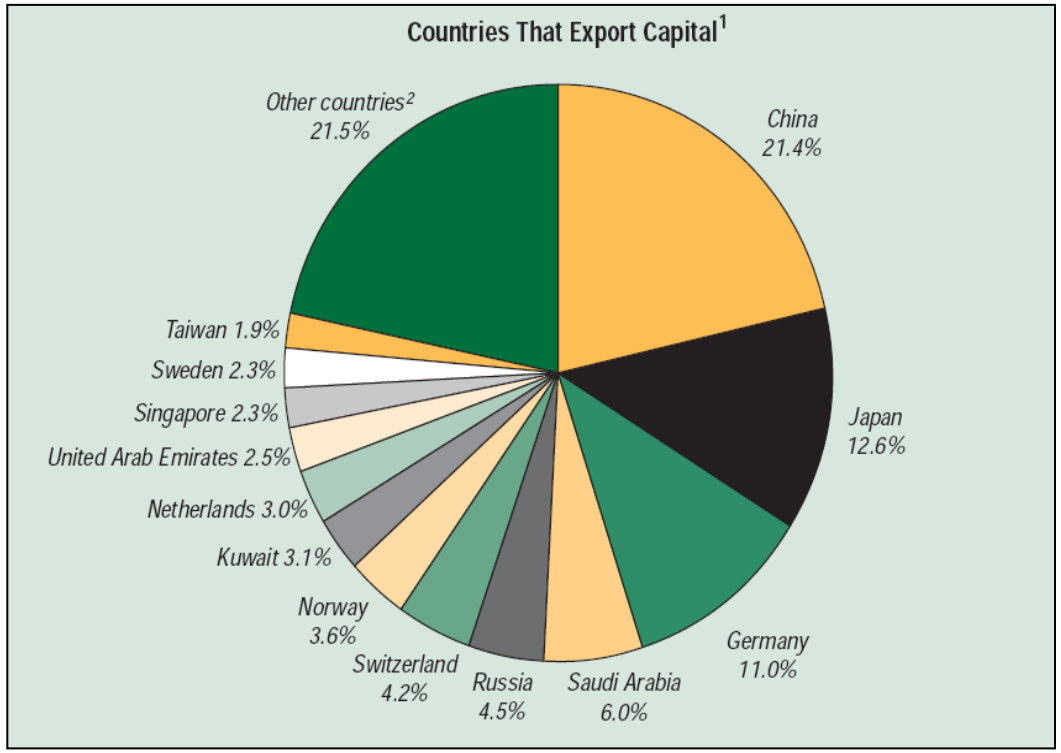
Balance of Payments of Switzerland

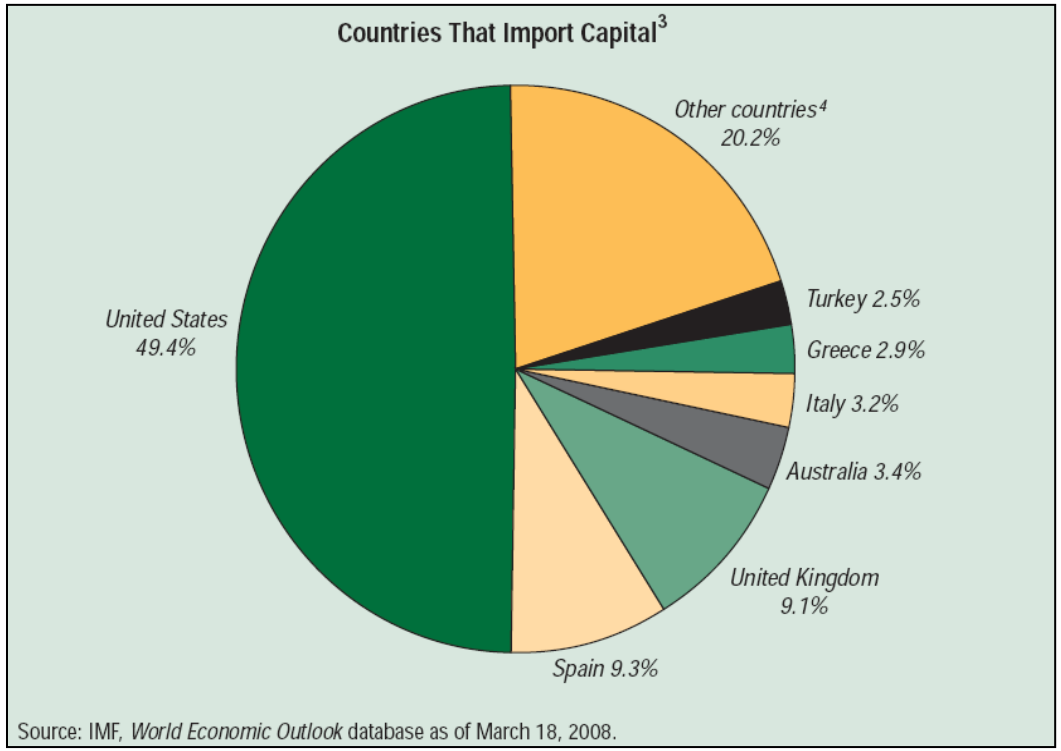


Source: SNB

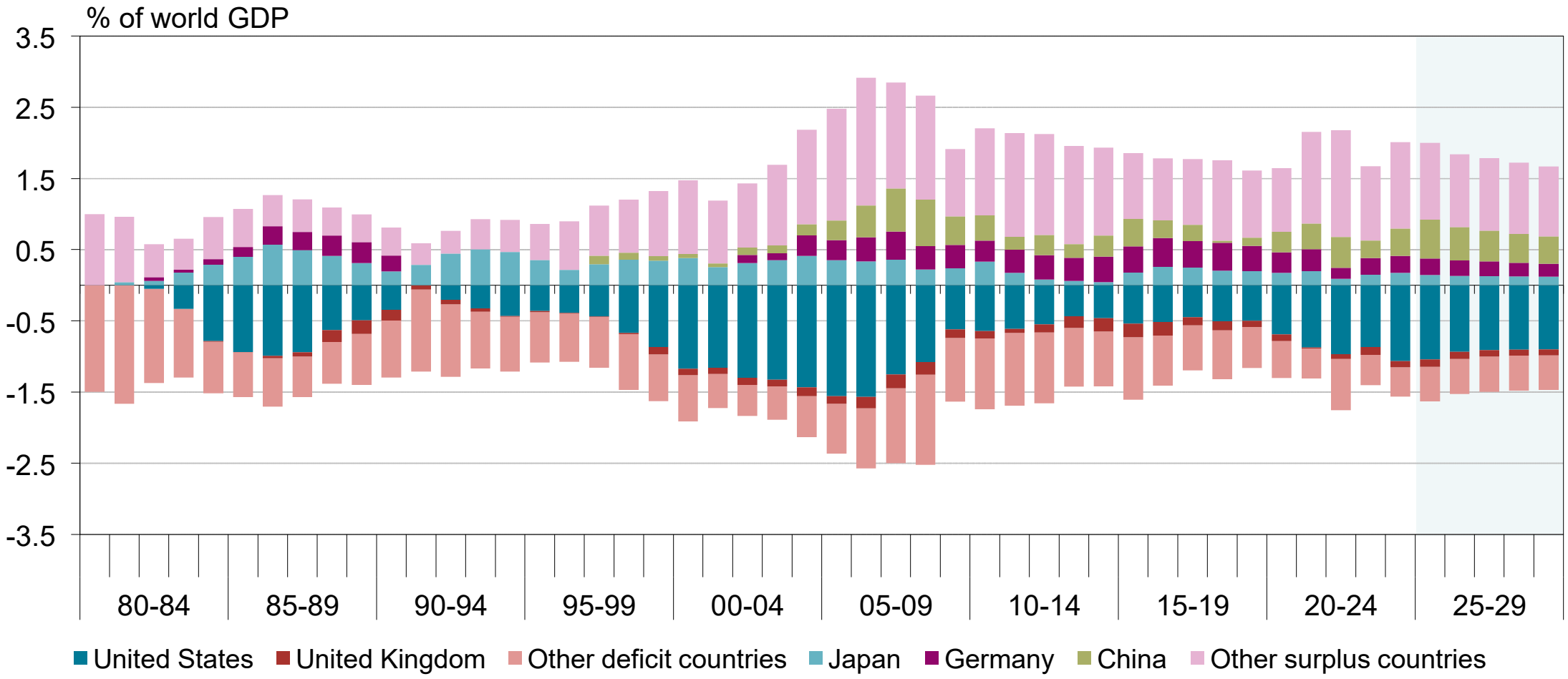
Global Financial Imbalances

- U.S. current account deficit in 2008 was as high as 6% of U.S. GDP!
- This was financed mostly by capital inflows from Asia and oil exporters.





Current account imbalances in the world

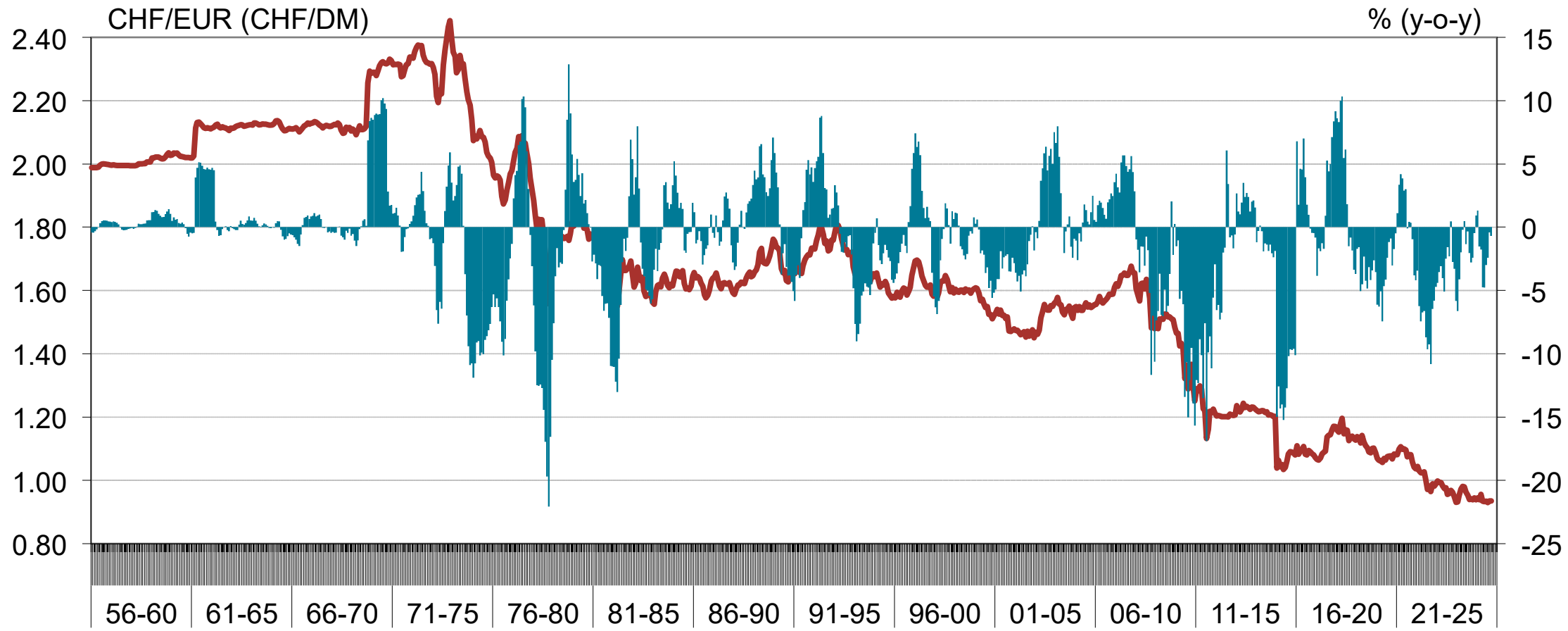


Source: IMF

The Prices for International Transactions: Real and Nominal Exchange Rates

- International transactions are influenced by international prices
- The two most important international prices are the nominal exchange rate and the real exchange rate
- The ***nominal exchange rate*** is the rate at which a person can trade the currency of one country for the currency of another and can be expressed in two ways:
  - In units of foreign currency per one Swiss Franc
  - And in units of Swiss Francs per one unit of the foreign currency
    - Assume the exchange rate between the Japanese yen and U.S. dollar is 80 yen to one dollar
      - One U.S. dollar trades for 80 yen, or one yen trades for 1/80 (= 0.0125) of a dollar
- ***Appreciation*** refers to an increase in the value of a currency as measured by the amount of foreign currency it can buy
- ***Depreciation*** refers to a decrease in the value of a currency as measured by the amount of foreign currency it can buy
  - If a Swiss Franc buys more foreign currency, there is an appreciation of the Swiss Franc
  - If it buys less there is a depreciation of the Swiss Franc

Nominal exchange rate Switzerland-Germany



Source: SNB

Real Exchange Rates

- The ***real exchange*** rate is the rate at which a person can trade the goods and services of one country for the goods and services of another
- The real exchange rate compares the prices of domestic goods and foreign goods in the domestic economy
  - If a case of Swiss beer is twice as expensive as German beer, the real exchange rate is 2 cases of German beer per case of Swiss beer.
- The real exchange rate depends on the nominal exchange rate (foreign currency / local currency = fc/lc) and the prices of goods in the two countries measured in local currencies
- The real exchange rate is a key determinant of how much a country exports and imports

$$\textit{Real exchange rate} = \frac{\textit{Nominal exchange rate } (fc/lc) \times \textit{Domestic price}}{\textit{Foreign price}}$$

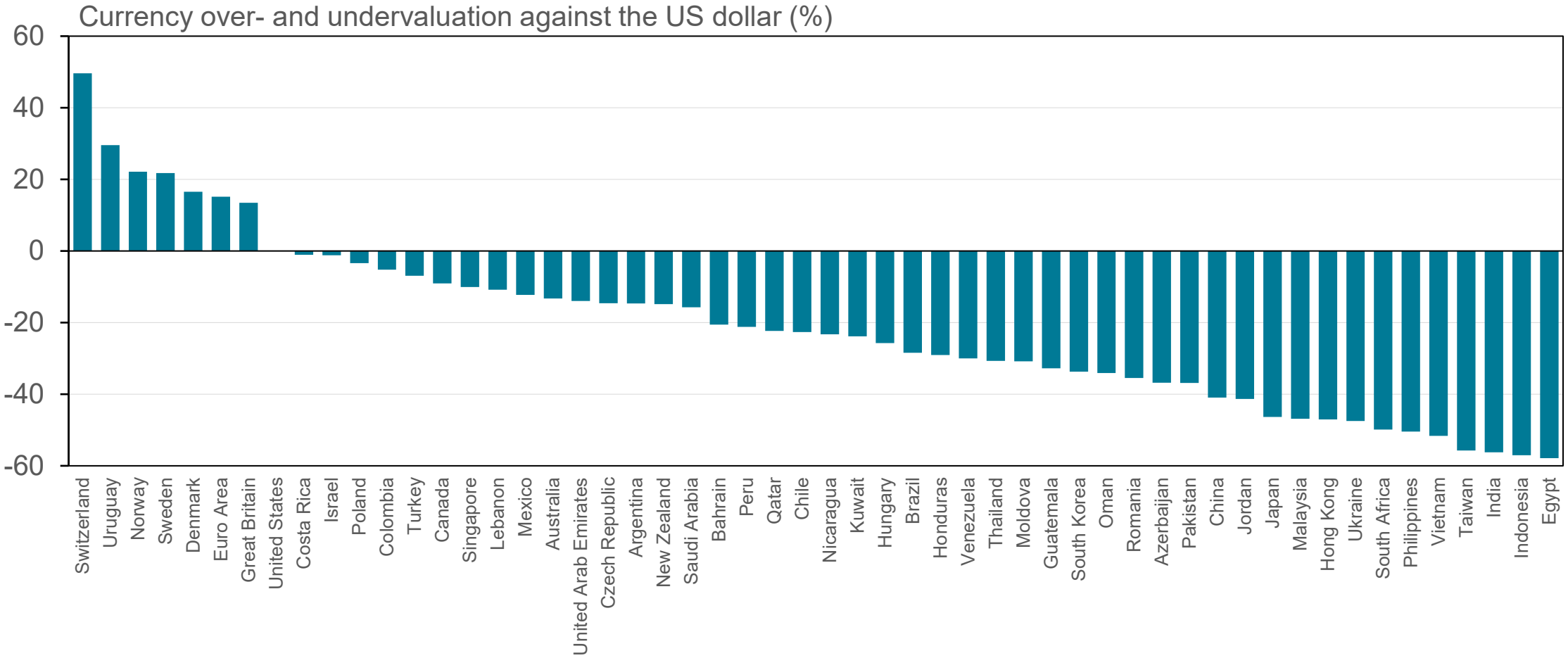
McZample

- one good: Big Mac
- price in the United States $P^* = \$\ 6.01$
- price in Switzerland: $P = \text{CHF}\ 7.20$
- nominal exchange rate: $e = 1.25\ \$/\text{CHF}$

- Real exchange rate: $\varepsilon = 1.25 * 7.20 / 6.01 = 1.50$
  - (Overvaluation of the Swiss France relative to the US Dollar: 50%)

To buy a Swiss Big Mac, someone from the US would have to pay an amount that could buy 1.5 Big Macs in the US

The Big-Mac-Index



Source: The Economist (July 2025)

The Big Mac Index (The Economist, July 2025)

- In America, a McDonald's Big Mac costs $6.01
- In Taiwan, the same burger costs just NT$ 78, equivalent to $2.66 at market exchange rates
  - Since NT$ 78 can buy as much burger as US$ 6.01, a US$ should be worth NT$ 12.98
  - In fact, a US$ costs NT$ 19.31, suggesting that the NT$ is undervalued by about 56%
- The euro was overvalued by 15% relative to the US dollar
- The Swiss franc is the most expensive currency (for buying a big mac)
  - It was overvalued by 50% relative to the US dollar in July 2025

Source: https://www.economist.com/big-mac-index

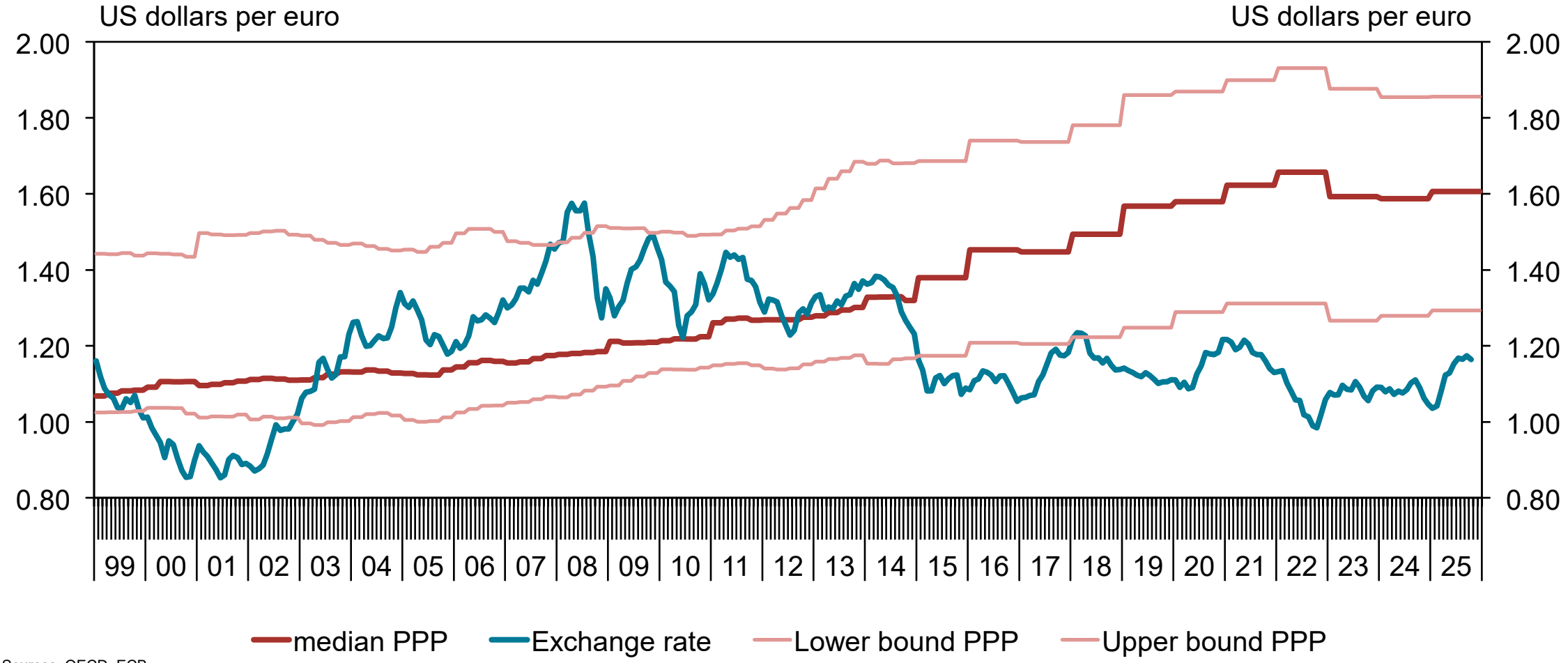
Real Exchange Rates

- A depreciation (fall) in the Swiss real exchange rate means that Swiss goods have become cheaper relative to foreign goods
- This encourages consumers both at home and abroad to buy more Swiss goods and fewer goods from other countries
- As a result, Swiss exports rise, and Swiss imports fall, and both of these changes raise Swiss net exports
- Conversely, an appreciation in the Swiss real exchange rate means that Swiss goods have become more expensive compared to foreign goods, so Swiss net exports fall

A First Theory of Exchange-Rate Determination: Purchasing-Power Parity

- The ***purchasing-power parity (PPP) theory*** is the simplest and most widely accepted theory explaining the variation of currency exchange rates
  - PPP is a theory of exchange rates whereby a unit of any given currency should be able to buy the same quantity of goods in all countries
- The PPP theory is based on a principle called the law of one price
  - According to the law of one price, a good must sell for the same price in all locations
  - If the law of one price were not true, unexploited profit opportunities would exist
    - The process of taking advantage of differences in prices in different markets is called arbitrage
      - If arbitrage occurs, eventually prices that differed in two markets would necessarily converge
- According to the PPP theory, a currency must have the same purchasing power in all countries and exchange rates move to ensure that
  - The nominal exchange rate between two currencies reflects the different price levels in the countries
- Limitations of the PPP theory
  - Many goods are not easily traded or shipped from one country to another
  - Tradable goods are not always perfect substitutes when they are produced in different countries

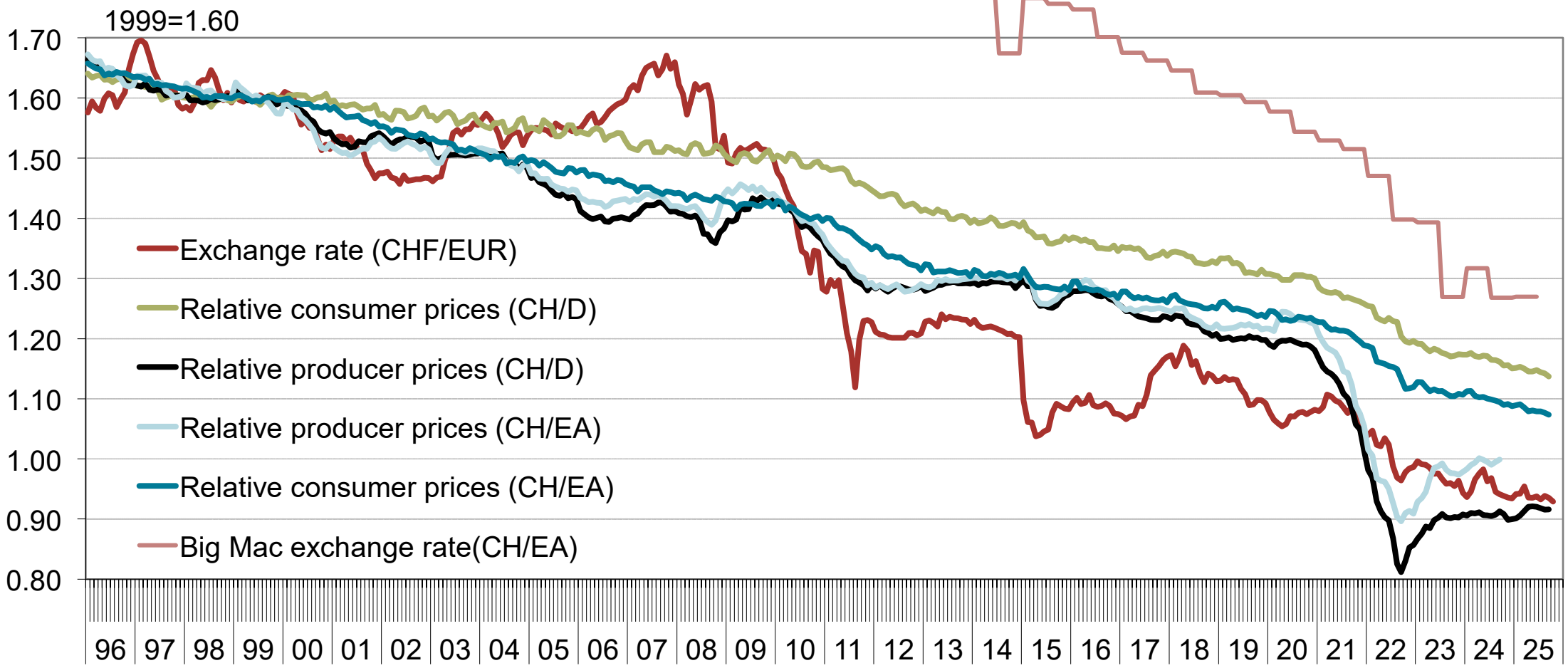
Exchange rate of the euro against the US dollar and PPP



Sources: OECD, ECB

Notes: The nominal exchange rate is based on monthly data, while the exchange rate based on purchasing power parity (PPP) is given at a quarterly frequency. The US dollar-euro PPP rate is calculated as the GDP-weighted average of the euro country-specific PPP estimates vis-à-vis the US dollar. The PPP upper bound represents the 95th percentile of the euro country-specific PPP estimates vis-à-vis the US dollar; the lower bound the 5th percentile. In calculating these bounds, the 11 euro area member countries with the largest GDP weights are used.

PPP and the CHF/EUR Exchange Rate



Sources: Eurostat, OECD, Macrobond

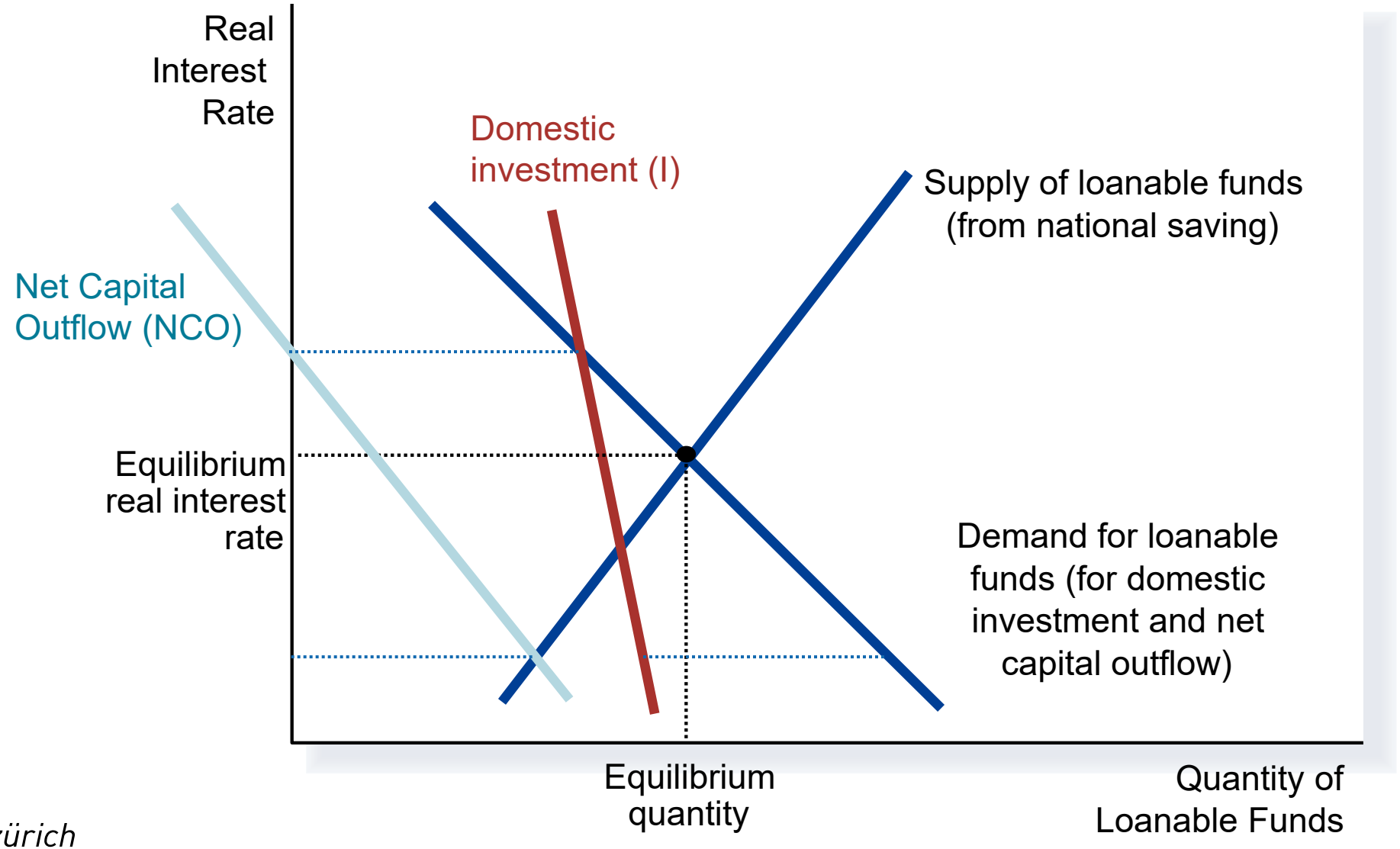
A Macroeconomic Theory of the Open Economy

- Key Macroeconomic Variables in an Open Economy
  - The important macroeconomic variables of an open economy include:
    - net exports
    - net foreign investment (= net capital outflow)
    - nominal exchange rates
    - real exchange rates
- Basic Assumptions of a Macroeconomic Model of an Open Economy
  - The model takes the economy's GDP as given
  - The model takes the economy's price level as given (nominal=real)
- To understand forces at work in an open economy, we must consider two markets:
  - The market for loanable funds
    - Co-ordinates savings, investment and net capital outflow
  - The market for foreign currency
    - Co-ordinates people who wish to exchange domestic to foreign currency

The Market for Loanable Funds

- The Market for Loanable Funds: $S = I + NCO$
- The supply of loanable funds comes from national saving (*S*)
  - A higher real interest rate encourages people to save, i.e. increase the supply of loanable funds
- The demand for loanable funds comes from domestic investment (*I*) and net capital outflows (*NCO*)
  - A higher real interest rate discourages firms to invest in new buildings, machines & equipment
  - A higher real interest rate discourages people to buy assets abroad
- The interest rate adjusts to bring supply and demand for loanable funds into balance
  - At the equilibrium interest rate, the amount that people want to save exactly balances the desired quantities of investment and net capital outflows

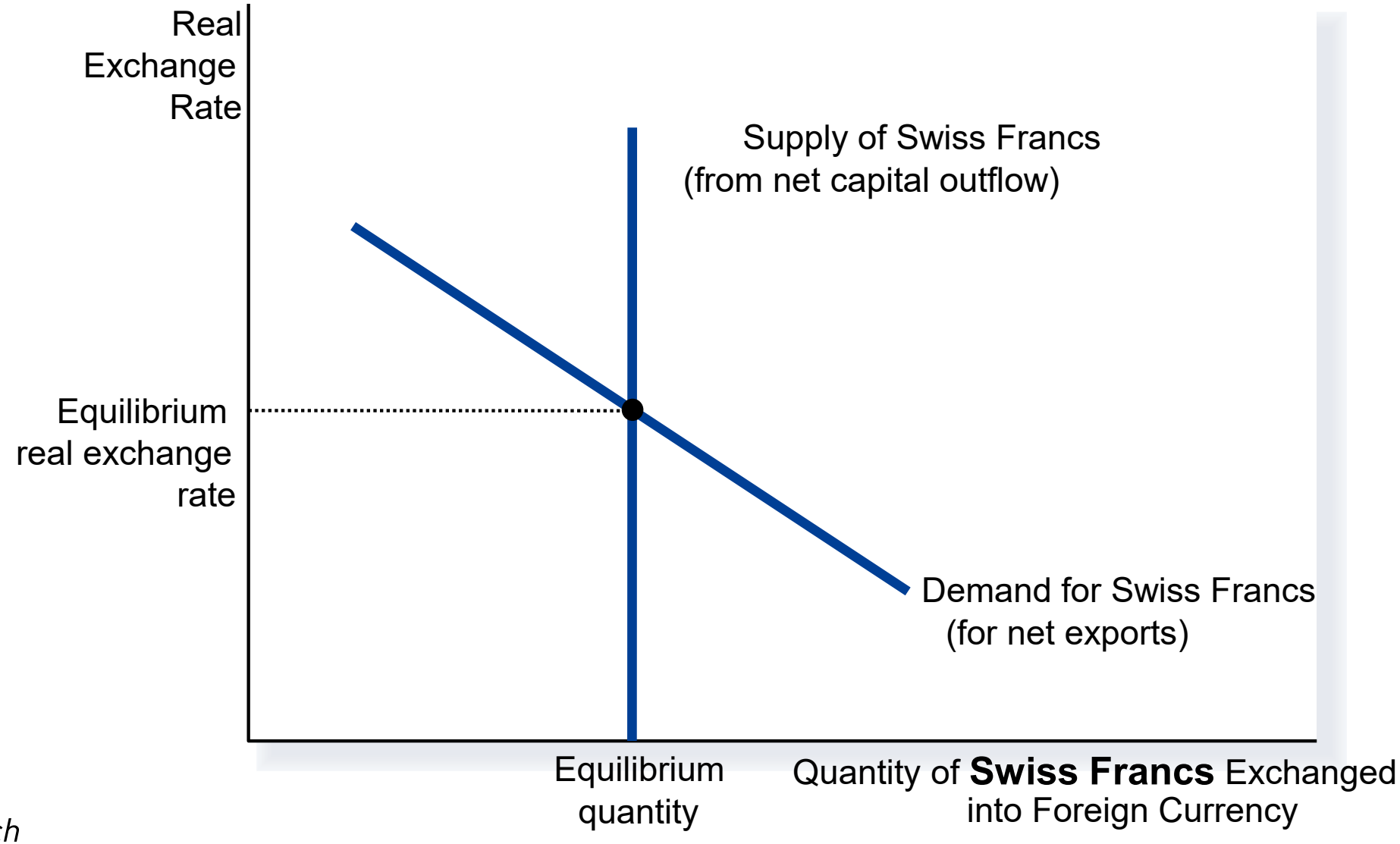
The Market for Loanable Funds



The Market for Foreign-Currency Exchange

- The two sides of the foreign currency exchange market are represented by NCO and NX
  - NCO (net capital outflow) represents the imbalance between the purchases and sales of assets
  - NX (net exports) represents the imbalance between exports and imports of goods and services
- For an economy as a whole, NCO and NX must balance, or: ***NCO = NX***
  - Supply of domestic currency = demand for domestic currency
- In the market for foreign-currency exchange, Swiss Francs are traded for foreign currencies
  - The price that balances the supply and demand for foreign-currency is the real exchange rate
  - A real appreciation of the Swiss Franc will be defined as a higher level of its real exchange rate
- A low real exchange rate of the Swiss Franc, leads to strong exports and a high demand for its currency
  - The demand curve (*NX*) for domestic currency is downward sloping
  - The supply curve (*NCO*) is vertical because the quantity of Swiss Francs supplied is unrelated to the (current) real exchange rate
    - It does depend upon the expected change in the real exchange rate during the investment horizon
    - We (implicitly) assume that the best forecast for the future real exchange rate is its current level
      - This implies that the (expected) change in the real exchange rate equals zero

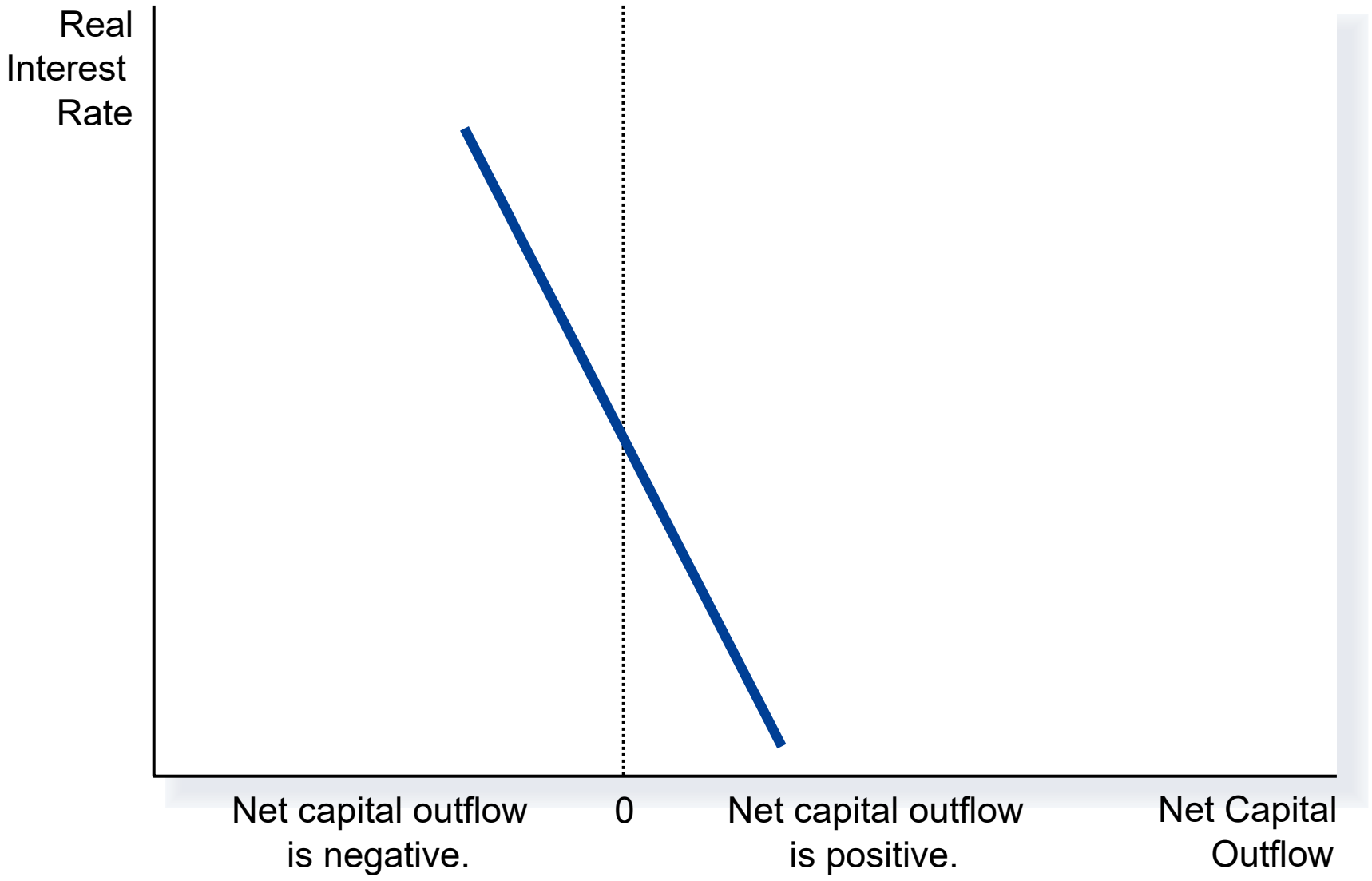
The Market for Foreign-Currency Exchange



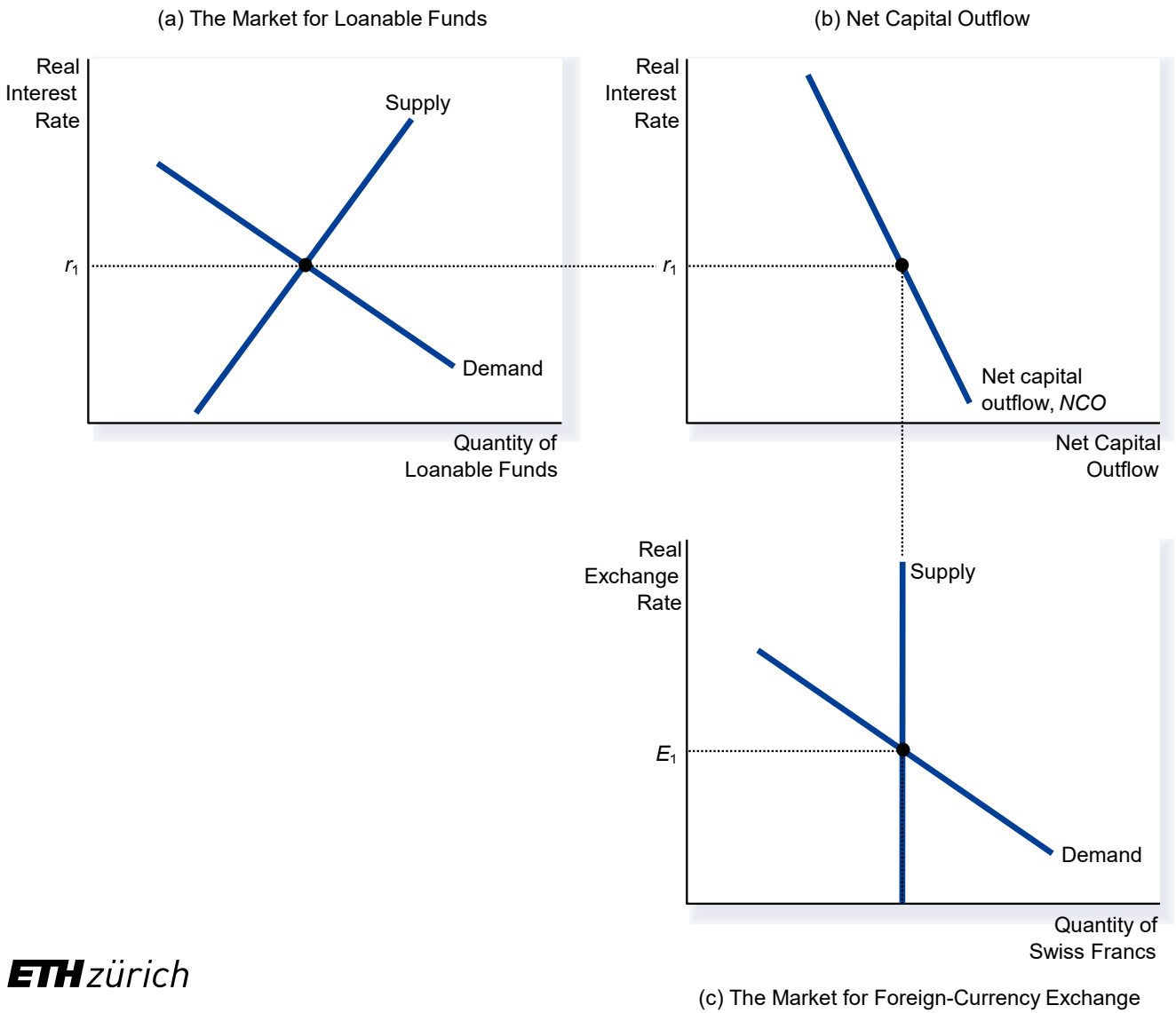
Equilibrium in the Open Macroeconomy

- The Market for Foreign-Currency Exchange
  - The real exchange rate adjusts to balance the supply and demand for Swiss Francs
    - At the equilibrium real exchange rate, the demand for Swiss Francs to buy net exports exactly balances the supply of Swiss Francs to be exchanged into foreign currency to buy assets abroad
- In the market for ***loanable funds***, supply comes from national saving and demand comes from domestic investment and net capital outflow.
- In the market for ***foreign currency exchange***, supply comes from net capital outflow and demand comes from net exports
- Equilibrium in the Open Economy
  - The market for loanable funds and the market for foreign exchange are related to each other
  - Net capital outflow links the loanable funds market and the foreign currency exchange market
  - The key determinant of net capital outflow is the real interest rate

How Net Capital Outflow Depends on the Interest Rate



Simultaneous Equilibrium in the Two Markets

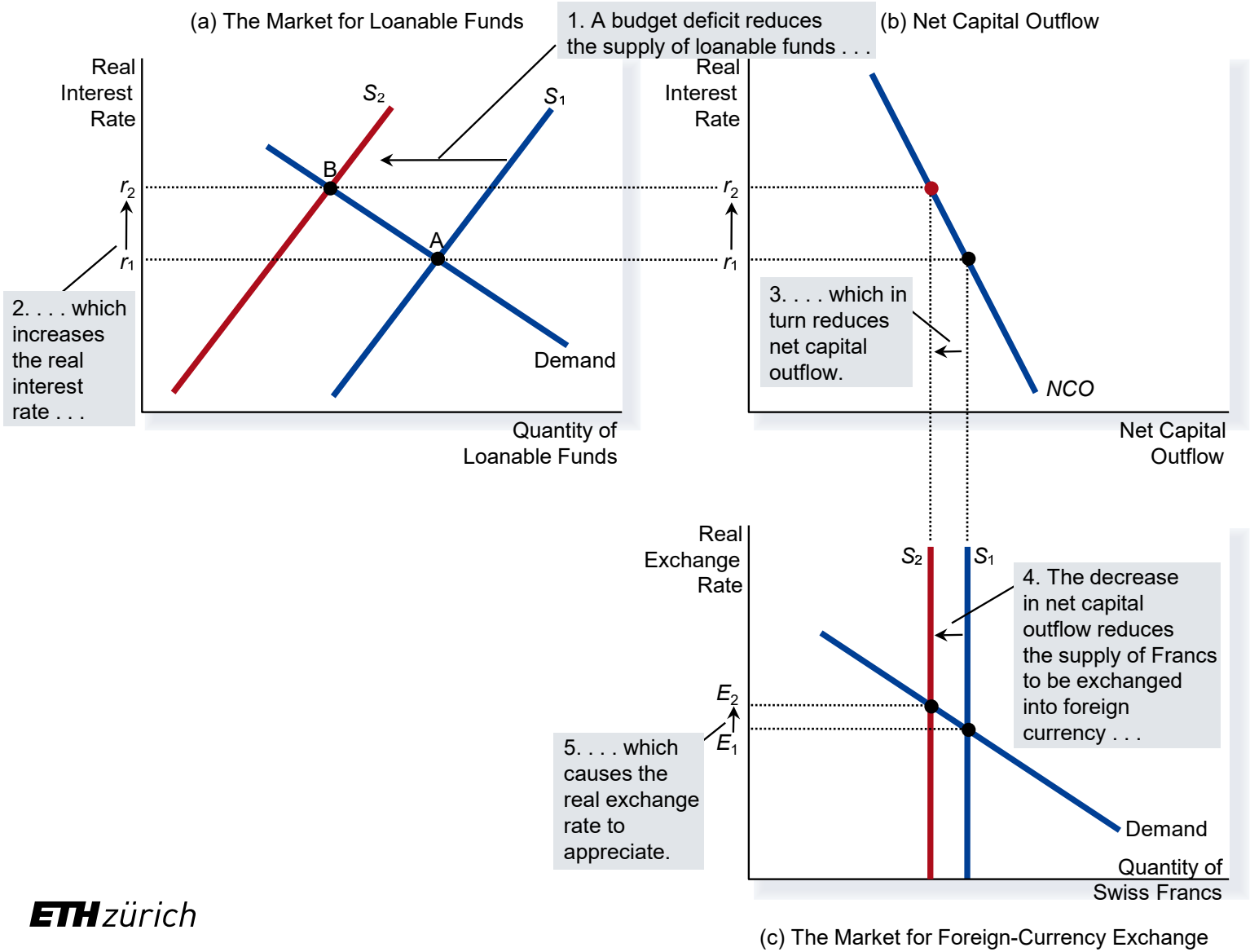


- Prices in the loanable funds market and the foreign-currency exchange market adjust simultaneously to balance supply and demand in these two markets
- As they do, they determine the macroeconomic variables of national saving, domestic investment, net foreign investment (net capital outflow), and net exports

How Policies and Events Affect an Open Economy

- The magnitude and variation in important macroeconomic variables depend on the following:
  - Government budget deficits
  - Trade policies
  - Political and economic stability

The Effects of Government Budget Deficit



- In an open economy, government budget deficits
  - reduce the supply of loanable funds,
  - drive up the interest rate,
  - crowd out domestic investment,
  - cause net foreign investment (net capital outflow) to fall
  - reduces the supply of Swiss Francs to be exchanged into foreign currency
  - Causes the real exchange rate to appreciate

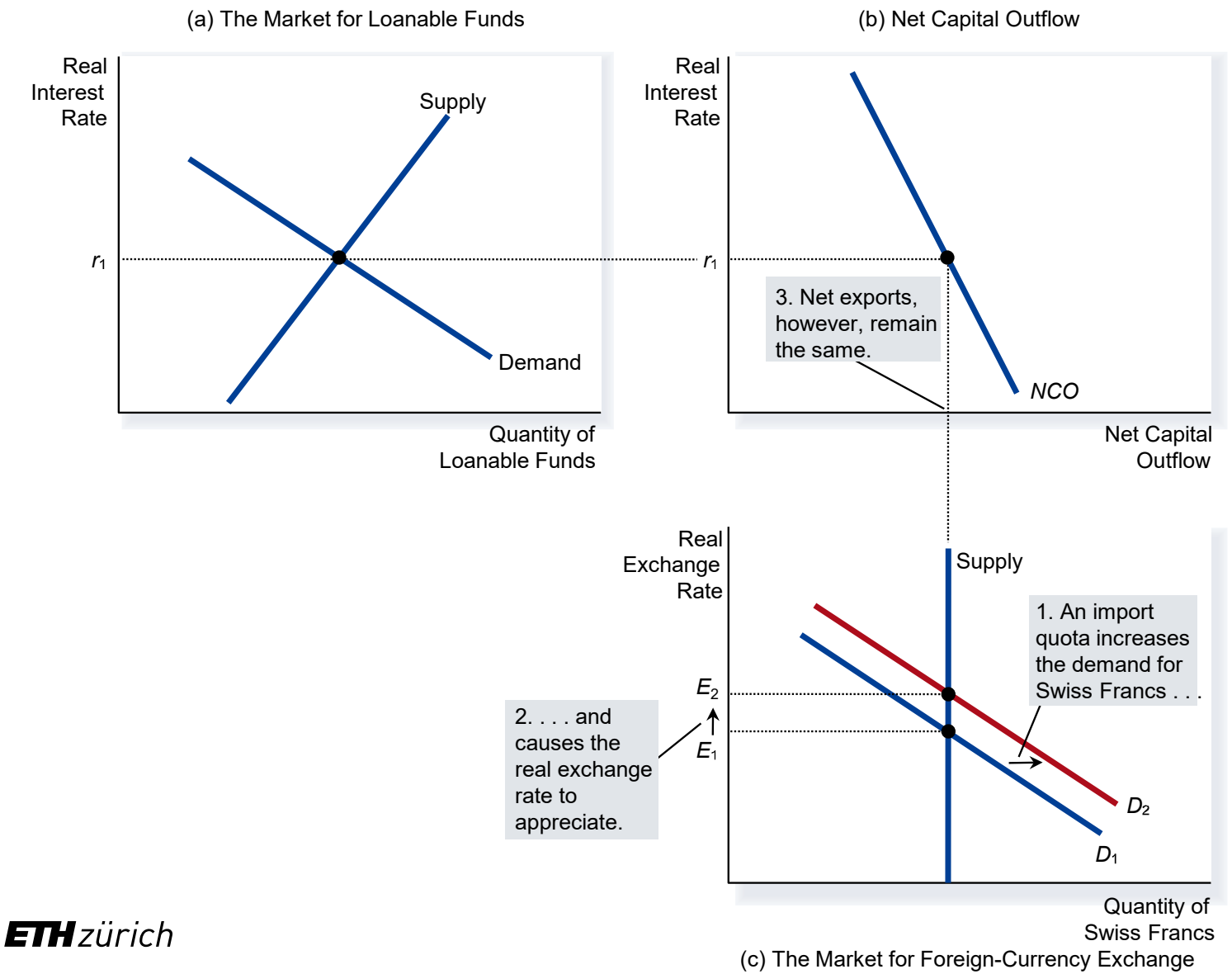
Trade Policy

- A trade policy is a government policy that directly influences the quantity of goods and services that a country imports or exports.
  - Tariff: A tax on an imported good
  - Import quota: A limit on the quantity of a good produced abroad and sold domestically
- Because they do not change national saving or domestic investment, trade policies do not affect the trade balance
  - For a given level of national saving and domestic investment, the real exchange rate adjusts to keep the trade balance the same
  - Trade policies have a greater effect on microeconomic than on macroeconomic markets

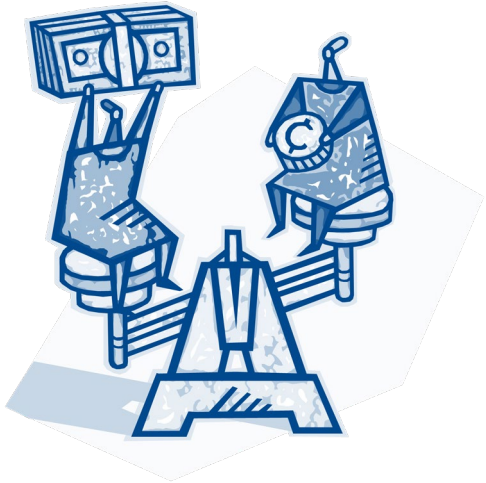
Trade Policy: Effect of an Import Quota

- Because foreigners need Swiss Francs to buy Swiss net exports, there is an increased demand for Swiss Francs in the market for foreign-currency
  - This leads to an appreciation of the real exchange rate
- There is no change in the interest rate because nothing happens in the loanable funds market
- There will be no change in net exports
- There is no change in net foreign investment even though an import quota reduces imports
- An appreciation of the Swiss Franc in the foreign exchange market encourages imports and discourages exports
- This offsets the initial increase in net exports due to the import quota

The Effects of an Import Quota



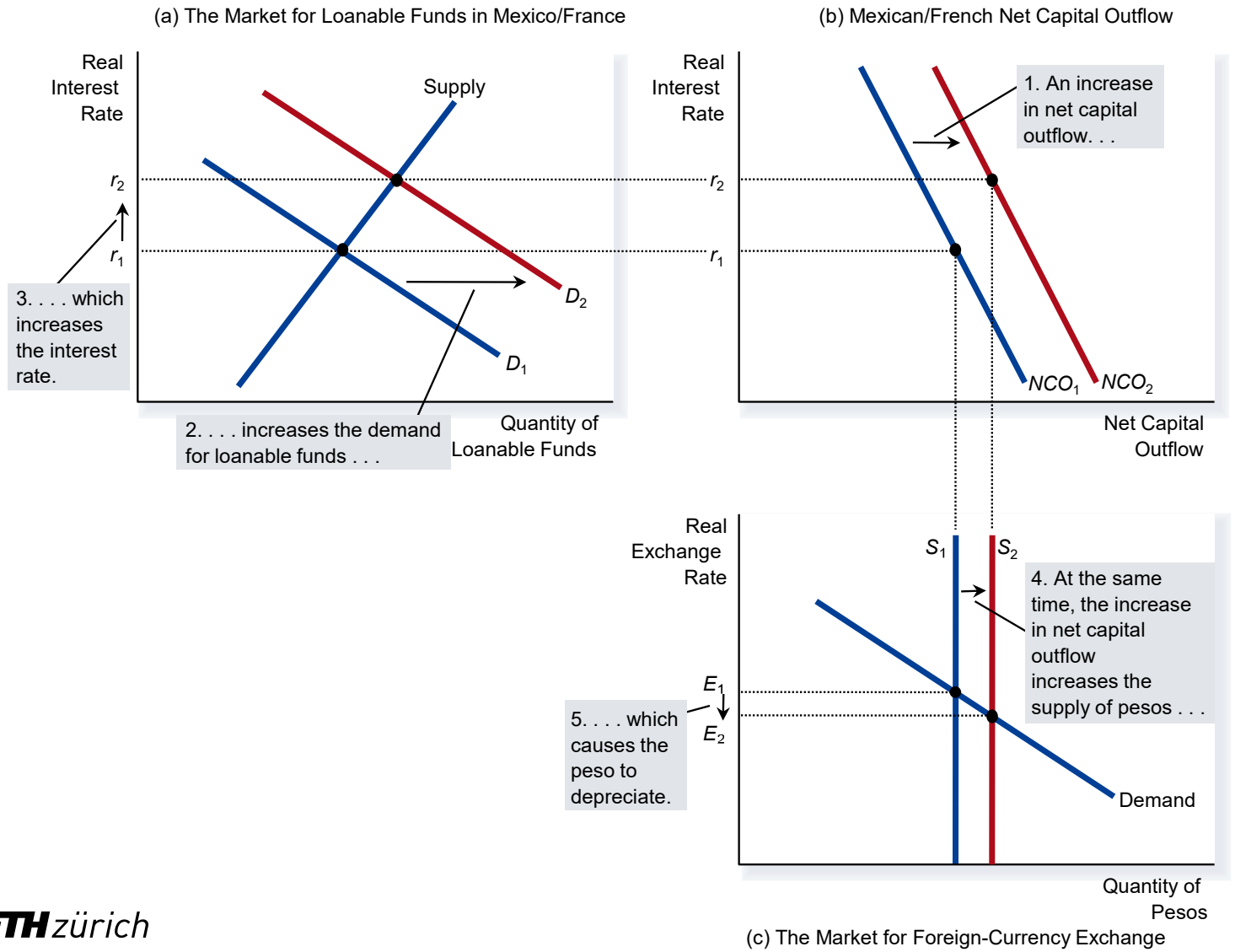
- Effect of an Import Quota
  - Trade policies do not affect the trade balance



Political Instability and Capital Flight

- ***Capital flight*** is a large and sudden reduction in the demand for assets located in a country
  - If investors become concerned about the safety of their investments, capital can quickly leave an economy
  - Interest rates increase and the domestic currency depreciates
- Capital flight has its largest impact on the country from which the capital is fleeing, but it also affects other countries
- When investors around the world observed political problems in Mexico in 1994, they sold some of their Mexican assets and used the proceeds to buy assets of other countries
  - This increased Mexican net capital outflow
    - The demand in the loanable funds market increased, which increased the interest rate
    - This increased the supply of pesos/euros in the foreign-currency exchange market, causing a depreciation

The Effects of Capital Flight



Summary

- Net exports are the value of domestic goods and services sold abroad minus the value of foreign goods and services sold domestically
- Net capital outflow is the acquisition of foreign assets by domestic residents minus the acquisition of domestic assets by foreigners
- An economy's net capital outflow always equals its net exports
- An economy's saving can be used to either finance investment at home or to buy assets abroad
- The nominal exchange rate is the relative price of the currency of two countries
- The real exchange rate is the relative price of the goods and services of two countries
- When the nominal exchange rate changes so that each Swiss Franc buys more foreign currency, the Swiss Franc is said to appreciate or strengthen
- When the nominal exchange rate changes so that each Swiss Franc buys less foreign currency, the Swiss Franc is said to depreciate or weaken

Summary

- According to the theory of purchasing-power parity, a unit of currency should buy the same quantity of goods in all countries
- The nominal exchange rate between the currencies of two countries should reflect the countries' price levels in those countries
- To analyze the macroeconomics of open economies, two markets are central—the market for loanable funds and the market for foreign-currency exchange
- In the market for loanable funds, the interest rate adjusts to balance supply for loanable funds (from national saving) and demand for loanable funds (from domestic investment and net capital outflow)
- In the market for foreign-currency exchange, the real exchange rate adjusts to balance the supply of Swiss Francs (for net capital outflow) and the demand for Swiss Francs (for net exports)
- Net capital outflow is the variable that connects the two markets
- A policy that reduces national saving, such as a government budget deficit, reduces the supply of loanable funds and drives up the interest rate
- The higher interest rate reduces net capital outflow, reducing the supply of Swiss Francs
- The Swiss Franc appreciates, and net exports fall